# Microsoft Azure Tenant

## Health Check Report

Prepared for <CUSTOMER>

**Author:** [Author]
**Date:** Click or tap here to enter text.
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## Preface

The information contained within this document will be treated with confidentiality and shall be only used by <COMPANY> for the purposes of this engagement.

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| 1.1 | 18/11/2021 | Hani Jafri | Added sections 11.2: Summary of Recommendations, 11.3: Service Improvement Plan |

### Distribution List

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# 1 Executive Summary

<PARTNER> recently approached <COMPANY> to explore the opportunity to deliver a Managed Service for <CUSTOMER> on their behalf.

As part of the service onboarding stage, it is expected that a health check will be performed across the existing Azure server environment, seeing the delivery of a detailed environment familiarisation exercise, along with cost, license and health analysis, to gain a better understanding of the underlying Azure services and resources in use within the current tenant along with ancillary components – such as identity and networking.

## 1.1 Document Purpose

Following the completion of the health check assessment, the purpose of this document is to outline the current state of the Azure tenant and highlight areas of improvement, for the following discovery elements:

- **Existing Azure tenant**
  - Azure tenant, governance and subscriptions
  - Azure cost monitoring / snapshot
  - Azure Active Directory (incl. identity synchronisation)
  - Azure networking services – To include hybrid connectivity health
  - Azure compute services – To include circa 30 servers
  - Azure database and storage services
  - Azure platform integration services
  - Azure service security and management (inc. Azure advisor)
  - Review existing Azure focused governance controls and policies


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# 2 Azure Governance

Azure governance is what defines the scaffolding for building everything else. Some key topics that should be covered here are naming and tagging convention, subscription model, policies, and compliance. Changing many of these things after resources have been deployed is time-consuming and introduces downtimes and risks – so it is very important to establish solid governance approach as early as possible. Another important consideration is flexibility – constant changes are intrinsic to a public cloud environment and it is vital to ensure that the governance approach will be able to accommodate and will speed up (rather than impede) expanding to the public cloud.

## 2.1 Azure Regions

### 2.1.1 Current State

| Azure Regions in use |
|---|
| Central US |
| East US |
| North Europe |
| Southeast Asia |
| West Europe |
| UK South |
| UK West |

### 2.1.2 Recommendations

| Azure Region Selection Recommendations | |
|---|---|
| Recommendation | Rationale |
| **Data Sovereignty**: Ensure the correct regions have been selected for residency and compliance needs. | For regulatory compliance considerations, data residency considerations may support or even mandate the physical locations where data can be stored, and how and when it can be transferred internationally. These regulations can differ significantly depending on jurisdiction. |
| **Disaster Recovery**: Identify primary and secondary regions for applications requiring regional resiliency. | If the primary region fails, a secondary region becomes crucial for business continuity. |
| **Performance**: Ensure resources are deployed in close proximity to target users. | There is global latency in how networks are distributed. Applications should be located close to target users for best performance. |
| **Pricing**: Factor cost into the decision-making process. | There is a price disparity between region with some regions offering lower costs per resource. |
| **Service Availability**: Ensure that the required Azure services are available in the selected regions. | Some Azure resources while under General Availability (GA) are not available in all regions. |
| | |


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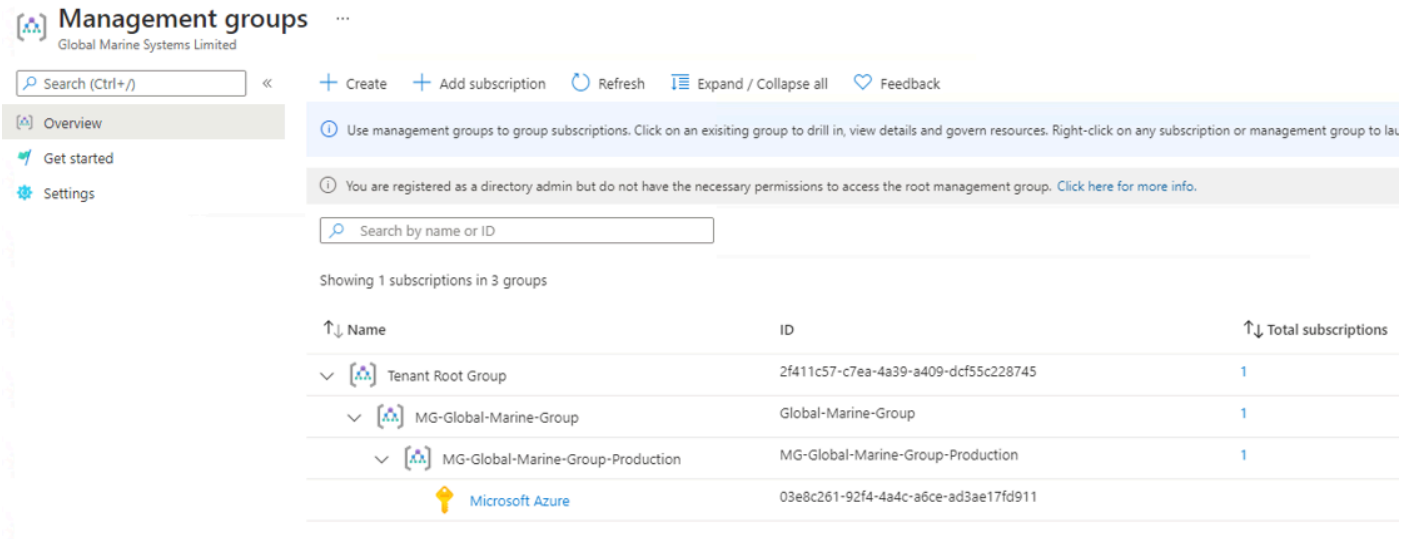
## 2.2 Management Groups

### 2.2.1 Current State




### 2.2.2 Recommendations

| Management Group Recommendations | |
|---|---|
| Recommendation | Rationale |
| Consider the creation of an additional Management Group for Non-production environments under the 'MG-Global-Marine-Group' Management Group. | Provides the flexibility to apply RBAC roles and Azure Policies at the Management Group level for Non-production and Production environments, to provide segregation of duties and separation of functional policies. |
| Enable 'Require write permission for creating new management groups' in Management Group settings. | Any user, by default, can create new management groups within a tenant. Admins of a tenant may wish to only provide these permissions to specific users to maintain consistency and conformity in the management group hierarchy. If enabled, a user requires the *Microsoft.Management/managementGroups/write* operation on the root management group to create new child management groups. |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## 2.3 Subscriptions

### 2.3.1 Current State

A single Subscription has been created for Production and Development environments.

| Subscriptions | | | |
|---|---|---|---|
| Subscription Name | Subscription ID | Subscription Plan | Directory |
| Microsoft Azure | 03e8c261-92f4-4a4c-a6ce-ad3ae17fd911 | Azure Plan | Global Marine Systems Limited (globalmarinesystems.onmicrosoft.com) |

### 2.3.2 Recommendations

| Subscription Recommendations | |
|---|---|
| Recommendation | Rationale |
| Consider the creation of an additional Subscription for non-production workloads. | Provides the flexibility to apply RBAC roles and Azure Policies at the Management Group level for Non-production and Production environments, to provide segregation of duties and separation of functional policies. |

## 2.4 Resource Groups

### 2.4.1 Current State

| Resource Groups | | |
|---|---|---|
| Resource Group Name | Subscription | Location |
| AzureBackupRG_southeastasia_1 | Microsoft Azure | Southeast Asia |
| AzureBackupRG_uksouth_1 | Microsoft Azure | UK South |
| cloud-shell-storage-northeurope | Microsoft Azure | North Europe |
| cloud-shell-storage-westeurope | Microsoft Azure | West Europe |
| Core-BootDiagStorageAccounts | Microsoft Azure | UK West |
| Core-DirectoryServices | Microsoft Azure | UK West |
| Core-DirectoryServices-ASR | Microsoft Azure | UK West |
| Core-Management | Microsoft Azure | UK West |
| Core-Networking | Microsoft Azure | UK West |
| core-networking-asr | Microsoft Azure | UK West |
| Dashboards | Microsoft Azure | UK South |
| DefaultResourceGroup-SEA | Microsoft Azure | Southeast Asia |
| DefaultResourceGroup-SUK | Microsoft Azure | UK South |
| FS03-DMZ | Microsoft Azure | UK South |
| gbazfsstorage | Microsoft Azure | UK South |


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| | | |
|---|---|---|
| GM-Azure-Storage | Microsoft Azure | UK South |
| HMNRecoveryService | Microsoft Azure | UK South |
| LogAnalyticsRG | Microsoft Azure | UK South |
| Meraki-UKS | Microsoft Azure | UK South |
| Meraki-UKSemtu7tifmoczk | Microsoft Azure | UK South |
| NetworkWatcherRG | Microsoft Azure | UK West |
| O365_Resources | Microsoft Azure | UK South |
| RG-Core-Monitoring | Microsoft Azure | North Europe |
| RG-EUS-PROD-LogAnalytics | Microsoft Azure | East US |
| RG-EUS-Production | Microsoft Azure | East US |
| RG-SEA-AKV | Microsoft Azure | Southeast Asia |
| RG-SEA-Azure-Migrate | Microsoft Azure | Southeast Asia |
| RG-SEA-PROD-LogAnalytics | Microsoft Azure | Southeast Asia |
| RG-SEA-PROD-MERAKI | Microsoft Azure | Southeast Asia |
| RG-SEA-PROD-MERAKI-MANAGED | Microsoft Azure | Southeast Asia |
| RG-SEA-PROD-SINGAPORE | Microsoft Azure | Southeast Asia |
| RG-SEA-PROD-TAIWAN | Microsoft Azure | Southeast Asia |
| RG-SEA-TEST | Microsoft Azure | Southeast Asia |
| RG-UKS-ASR | Microsoft Azure | UK South |
| RG-UKS-AZURE-DIGITAL-TWINS | Microsoft Azure | UK South |
| RG-UKS-Azure-Migrate | Microsoft Azure | UK South |
| RG-UKS-AZURE-MIGRATE-SQL | Microsoft Azure | UK South |
| RG-UKS-AZURE-SQL-MI | Microsoft Azure | UK South |
| RG-UKS-CHECKPOINT-CLOUDGUARD | Microsoft Azure | UK South |
| RG-UKS-DEV-Azure-Backups | Microsoft Azure | UK South |
| RG-UKS-DEV-O365 | Microsoft Azure | UK South |
| RG-UKS-DEV-SSH | Microsoft Azure | UK South |
| RG-UKS-DEV-TEST | Microsoft Azure | UK South |
| RG-UKS-DEV-WVD-FULL | Microsoft Azure | UK South |
| RG-UKS-IFS-P6 | Microsoft Azure | UK South |
| RG-UKS-PROD-ABERDEEN-STORAGE | Microsoft Azure | UK South |
| RG-UKS-PROD-AKV | Microsoft Azure | UK South |
| RG-UKS-PROD-FILE-SYNC | Microsoft Azure | UK South |
| RG-UKS-PROD-LogAnalytics | Microsoft Azure | UK South |
| RG-UKS-PROD-ROVCO-CWIND | Microsoft Azure | UK South |
| RG-UKS-PROD-WVD-IT | Microsoft Azure | UK South |
| RG-UKS-PURVIEW | Microsoft Azure | UK South |
| RG-UKS-SECURITY-CENTRE | Microsoft Azure | UK South |
| RG-UKS-STATIC-WEB-APP | Microsoft Azure | UK South |
| RG-UKS-WATCHGUARD-DEV | Microsoft Azure | UK South |
| RG-UKS-WATCHGUARD-FIREBOX-CLOUD | Microsoft Azure | UK South |


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| | | |
|---|---|---|
| RG-UKS-WATCHGUARD-FIREBOX-CLOUD-STORAGE | Microsoft Azure | UK South |
| RG-UKS-WEBAPP | Microsoft Azure | UK South |
| RG-UKS-WEBAPP-SCOTNIPAC | Microsoft Azure | UK South |
| RG-UKS-WVD-IT | Microsoft Azure | UK South |
| RG-UKS-WVD-OceanIQ | Microsoft Azure | UK South |
| RG-UKW-AKV | Microsoft Azure | UK West |
| RG-UKW-ASR | Microsoft Azure | UK West |
| Services-Exchange | Microsoft Azure | UK South |

### 2.4.2 Recommendations

| **Resource Group Recommendations** | |
|---|---|
| Recommendation | Rationale |
| Use resource groups to logically group resources together which share the same lifecycle. | Resources are deployed, updated, and deleted together. If one resource, such as a server, needs to exist on a different deployment cycle it should be in another resource group. |
| Review and delete empty resource groups:<br>• FS03-DMZ<br>• gbazfsstorage<br>• HMNRecoveryService<br>• RG-SEA-PROD-TAIWAN<br>• RG-SEA-TEST<br>• RG-UKS-ASR<br>• RG-UKS-DEV-Azure-Backups<br>• RG-UKS-DEV-WVD-FULL<br>• RG-UKS-PURVIEW<br>• RG-UKS-STATIC-WEB-APP<br>• RG-UKW-ASR | Prevents cloud sprawl and improves management of resources |
| Ensure resource locks are set for resource groups containing mission critical Azure resources. | Prevents against accidental deletion of resources. |

## 2.5 Naming Standards

### 2.5.1 Current State

| **Current Naming Standards** | | |
|---|---|---|
| **Naming Convention Prefixes** | | |
| Resource Type | Value | Notes |
| Virtual Machine | AZ<REGION><ENVIRONMENT><SERVICE><NODE> | Used to identify Azure virtual machines |


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| | | |
|---|---|---|
| Resource Groups | RG-<REGION>-<ENVIRONMENT>-<SERVICE> | Used to identify Azure Resource Groups |
| Other resources | LB-<REGION>-<ENVIRONMENT>-<SERVICE> | Used to identify all other resource types e.g. LB for Load Balancer |
| **GMG Azure Regions** | | |
| Region | Value | Notes |
| UK South | <PREFIX>UKS | Used to identify resources in UK South |
| UK West | <PREFIX>UKW | Used to identify resources in UK West |
| East US | <PREFIX>EUS | Used to identify resources in East US |
| South-East Asia | <PREFIX>SEA | Used to identify resources in South-East Asia |
| **Azure Environments** | | |
| Environment Name | Value | Notes |
| Production | <PREFIX><REGION>PROD | Used to identify resources in Production |
| Development | <PREFIX>DEV | Used to identify resources in Development |
| **Service** | | |
| Service Name | Value | Notes |
| E.g. Exchange | <ENVIRONMENT>EXC | Used to identify service or application name |
| E.g. Meraki | <ENVIRONMENT>MER | Used to identify service or application name |
| **Note**: Service or application names longer than three letters can be truncated to the first three characters or any recognisable abbreviation e.g. Exchange can be truncated to EXC. | | |
| **Resource Digits** | | |
| Resource ID Digit | Value | Notes |
| E.g. 01,02,03 | <SERVICE>01 | Used to identify a double-digit resource ID |

Examples:

- RG-UKS-PROD-FTP – a production resource group in UK South for hosting FTP services
- AZUKSPRODEXC01 – a production exchange server in UK South region
- AZSEADEVDUO01 – a development Duo application server in South-East Asia region
- AZUKWPRODDOC01 – a production domain controller in UK West region
- LB-UKS-PROD-NLB1- a production load-balancer in UK South region
- FW-SEA-PROD-FW1 – a production firewall in South-East Asia region

### 2.5.2 Recommendations

| **Naming Standards Recommendations** | |
|---|---|
| Recommendation | Rationale |
| Ensure all resources are named in adherence to naming standards | Consistent naming of resources is crucial to identifying resources not only in the Azure portal and in bills, but also in scripts that support Azure DevOps automation/orchestration for release management. |


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| | |
|---|---|
| Use recommended abbreviations for Azure resource types as prefixes in resource names: https://docs.microsoft.com/en-us/azure/cloud-adoption-framework/ready/azure-best-practices/resource-abbreviations | Using the recommended abbreviations for Azure resource types in resource names that represents the type of the Azure resource makes it easier to visually recognize application or service components. |
| Consider enforcing or auditing naming standards using custom Azure policies | Consistent naming of resources is crucial to identifying resources not only in the Azure portal and in bills, but also in scripts that support Azure DevOps automation/orchestration for release management. |
| Document naming rules and restrictions as part of the naming standards | Resource names can vary in length limits, valid characters and letter case types. Balancing the context embedded in a name with its scope and rules is important when develop naming standards. |

## 2.6 Tagging Standards

### 2.6.1 Current State

The following tags are required as a minimum when creating a resource group in the GMG cloud:

| **Tagging Standards Current State** | |
|---|---|
| Tag | Value |
| Owner | <Owner's Full Name> |
| Department | <Department Name e.g. IT> |
| Service | <Service or Application Name> |
| Environment | <Production or Dev> |
| Budget | <Budget Code> |
| Created | <Date Created e.g. 20/08/20)> |
| Expires | <Expiry Date if known> |

Tags are applied using the following policy initiatives:

| **Tagging Policies** | | |
|---|---|---|
| Policy Initiative Name | Policy Initiative Effect | Policy Scope |
| Append a tag and its value from the resource group | Append | Subscription: *Microsoft Azure* |
| Require tags on resource groups | Deny | Subscription: *Microsoft Azure* |

### 2.6.2 Recommendations

| **Tagging Standards Recommendations** | |
|---|---|
| Recommendation | Rationale |


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| | |
|---|---|
| Consider creating an allowed set of tag values for *Department*, *Service*, *Environment* and *Budget* tags by using the policy 'Require a tag and its value on resources' | Using an allowed set of tag values will ensure valid tag values are applied to all resources. For example, the VM *AZSEAINDOC02* has the tag *Service* with value *IaaS/PaaS*. Using an allowed set of tag values for the *Service* tag based on a list of defined services will allow for easier identification of the resource, for example, tag value *Active Directory Domain Services*. |

## 2.7 Azure Policies

### 2.7.1 Current State

| **Assigned Policies** | | |
|---|---|---|
| Assignment Name | Policy Type | Policy Scope |
| A security contact email address should be provided for your subscription | Policy | Management Group: *MG-Global-Marine-Group* |
| ASC DataProtection | Initiative | Subscription: *Microsoft Azure* |
| ASC Default | Initiative | Subscription: *Microsoft Azure* |
| Allowed virtual machine SKUs | Initiative | Subscription: *Microsoft Azure* |
| Append a tag and its value from the resource group | Initiative | Subscription: *Microsoft Azure* |
| Azure Defender for servers should be enabled | Policy | Subscription: *Microsoft Azure* |
| CIS Microsoft Azure Foundations Benchmark 1.1.0 | Initiative | Subscription: *Microsoft Azure* |
| Configure Azure Defender to be enabled on SQL servers | Policy | Subscription: *Microsoft Azure* |
| Deploy Advanced Threat Protection on Storage Accounts | Initiative | Subscription: *Microsoft Azure* |
| Deploy Advanced Threat Protection on Storage Accounts | Policy | Management Group: *MG-Global-Marine-Group* |
| Enforce Hybrid Use Benefit on Windows Servers | Initiative | Subscription: *Microsoft Azure* |
| Require tags on resource groups | Initiative | Subscription: *Microsoft Azure* |

The following policy exemptions have been created:


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| Total exemptions | Exemptions approaching expiration | Exemptions expired |
|---|---|---|
| 8 | 0 (within next 7 days) | 0 |

To create a new exemption, please view the assignment or the compliance.

| Policy exemption | Assignment | Scope | Exemption category | Expiration date |
|---|---|---|---|---|
| Microsoft Azure - Allowed virtual machine SKUs | Allowed virtual machine SKUs | Microsoft Azure | Waiver | -- |
| Disable MFA on Owner Accounts-Builtin | ASC Default (subscription: 03e8c261-92f4-4a4c-a6ce-ad3ae17fd911) | Microsoft Azure | Mitigated | -- |
| Allow Support for external Owners-Builtin | ASC Default (subscription: 03e8c261-92f4-4a4c-a6ce-ad3ae17fd911) | Microsoft Azure | waiver | -- |
| Allow more than 3 Owners on Subscription-Builtin | ASC Default (subscription: 03e8c261-92f4-4a4c-a6ce-ad3ae17fd911) | Microsoft Azure | waiver | -- |
| ASC-MFA should be enabled on accounts with write permissions o... | CIS Microsoft Azure Foundations Benchmark 1.1.0 | Microsoft Azure | Mitigated | -- |
| ASC-MFA should be enabled on accounts with write permissions o... | ASC Default (subscription: 03e8c261-92f4-4a4c-a6ce-ad3ae17fd911) | Microsoft Azure | Mitigated | -- |
| ASC-MFA should be enabled on accounts with read permissions on... | CIS Microsoft Azure Foundations Benchmark 1.1.0 | Microsoft Azure | Mitigated | -- |
| ASC-MFA should be enabled on accounts with read permissions on... | ASC Default (subscription: 03e8c261-92f4-4a4c-a6ce-ad3ae17fd911) | Microsoft Azure | Mitigated | -- |

The overall resource compliance is currently 71% as per below:

Scope: Microsoft Azure

Get notified of compliance state changes! Use event-based architecture to react to notifications with an Azure Function, Logic App, or any other supported event handler. Learn more https://aka.ms/policyPlusEventGrid

Overall resource compliance: 71% (509 out of 715)

Resources by compliance state: 715 — 509 - Compliant; 0 - Exempt; 206 - Non-compli...

Non-compliant initiatives: 4 out of 8

Non-compliant policies: 109 out of 325

LEARN MORE: Learn about Policy; Onboarding tutorial

| Name | Scope | Compliance state | Resource compliance | Non-Compliant Resources | Non-compliant policies |
|---|---|---|---|---|---|
| CIS Microsoft Azure Foundations Benchmark 1.1.0 | Microsoft Azure | Non-compliant | 45% (143 out of 321) | 178 | 42 |
| ASC Default (subscription: 03e8c261-92f4-4a4c-a6ce... | Microsoft Azure | Non-compliant | 8% (9 out of 112) | 103 | 53 |
| Append a tag and its value from the resource group | Microsoft Azure | Non-compliant | 98% (595 out of 609) | 14 | 7 |
| Require tags on resource groups | Microsoft Azure | Non-compliant | 94% (60 out of 64) | 4 | 7 |
| Enforce Hybrid Use Benefit on Windows Servers | Microsoft Azure | Compliant | 100% (33 out of 33) | 0 | 0 |

### 2.7.2 Recommendations

| Azure Policies Recommendations | |
|---|---|
| **Recommendation** | **Rationale** |
| Create the following policies:<br>• Allowed locations<br>• Allowed locations for resource groups<br>• Audit resource location matches resource group location<br>• Audit VMs that do not use managed disks<br>• Automation account variables should be encrypted | The listed policies are an example of common recommended policies to enforce governance for resource consistency, regulatory compliance, security, cost, and management across the Azure environment. |


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| | |
|---|---|
| • Configure Azure Defender for DNS to be enabled<br>• Configure Azure Defender for Key Vaults to be enabled<br>• Configure Azure Defender for Resource Manager to be enabled<br>• Configure Azure Defender for Storage to be enabled<br>• Configure Azure Defender for servers to be enabled<br>• Custom subscription owner roles should not exist<br>• Enable Azure Monitor for VMs<br>• RDP access from the Internet should be blocked<br>• SSH access from the Internet should be blocked | |
| Review non-compliant policies and execute remediation tasks to put resources in to a compliant state. | Non-compliant resources will not adhere to organizational governance standard. |
| Review policy exemption for *Allowed Virtual Machine SKUs*, update corresponding policy assignment and remove exemption | The policy assignment should be updated to reflect any additional Virtual Machine SKUs required to avoid having to create an exemption |
| Use policy exemptions sparingly and add ensure each exemption has sufficient justification | Creating too many policy exemptions without justification can result in non-compliance of Azure resources |

## 2.8 Cost Management

### 2.8.1 Current State

| **Cost Management Features** | |
|---|---|
| Feature Name | Status |
| Azure Hybrid Use Benefit on SQL Servers | Not enabled |
| Azure Hybrid Use Benefit on Windows Servers | Enabled via Azure Policy |
| Azure VM Reserved Instances | Not enabled |
| Subscription Budgets and Alerts | Enabled |
| Tags for cost reporting | Enabled – refer to tagging section |

Subscription budgets and alerts have been created and scoped at the resource group level as follows:


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| Name | Scope | Reset period | Creation date | Expiration date | Budget | Forecasted | Evaluated spend | Progress |
|---|---|---|---|---|---|---|---|---|
| AB-UKS-WEBAPP-SCOTNIPAC | RG-UKS-WEBAPP-SCOTNIPA... | Monthly | 7/1/2020 | 6/30/2022 | £60.00 | | £0.00 | 0.00% |
| AB-AZURE-PRODUCTION | 03e8c261-92f4-4a4c-a6ce-a... | Monthly | 7/1/2020 | 6/30/2022 | £20,000.00 | | £16,503 | 82.51% |
| AB-UKS-PROD-ABERDEEN-STOR... | RG-UKS-PROD-ABERDEEN-S... | Monthly | 7/1/2020 | 6/30/2022 | £200.00 | | £314.56 | 100.00% |
| AB-GM-AZURE-STORAGE | GM-Azure-Storage (Resourc... | Monthly | 7/1/2020 | 6/30/2022 | £3,000.00 | | £2,256 | 75.21% |
| AB-UKS-PROD-BARRACUDA | RG-UKS-PROD-BARRACUDA... | Monthly | 9/1/2020 | 8/31/2022 | £500.00 | | £0.00 | 0.00% |
| AB-SEA-PROD-SINGAPORE | RG-SEA-PROD-SINGAPORE (... | Monthly | 9/1/2020 | 8/31/2022 | £500.00 | | £236.95 | 47.39% |
| AB-UKS-PROD-ROVCO-CWIND | RG-UKS-PROD-ROVCO-CWI... | Monthly | 9/1/2020 | 8/31/2022 | £200.00 | | £67.64 | 33.82% |
| AB-UKS-EXPRESSROUTE | Core-Networking (Resource ... | Monthly | 9/1/2020 | 8/31/2022 | £2,500.00 | | £1,761 | 70.46% |
| RG-UKS-DEV-WVD | RG-UKS-DEV-WVD (Resourc... | Monthly | 9/1/2020 | 8/31/2022 | £100.00 | | £0.00 | 0.00% |
| RG-SEA-AZURE-MIGRATE | RG-SEA-Azure-Migrate (Res... | Monthly | 10/1/2020 | 9/30/2022 | £100.00 | | £32.03 | 32.03% |
| AB-UKS-AZURE-MIGRATE | RG-UKS-Azure-Migrate (Res... | Monthly | 10/1/2020 | 9/30/2022 | £100.00 | | £8.30 | 8.30% |
| AB-UKS-AZURE-MIGRATE-SQL | RG-UKS-AZURE-MIGRATE-S... | Monthly | 10/1/2020 | 9/30/2022 | £300.00 | | £6.98 | 2.33% |
| AB-UKS-AZURE-SQL-MI | RG-UKS-AZURE-SQL-MI (Re... | Monthly | 10/1/2020 | 9/30/2022 | £300.00 | | £677.44 | 100.00% |
| AB-UKS-DEV-TEST | RG-UKS-DEV-TEST (Resourc... | Monthly | 12/1/2020 | 11/30/2022 | £200.00 | | £0.03 | 0.02% |
| AB-SEA-PROD-TAIWAN | RG-SEA-PROD-TAIWAN (Res... | Monthly | 12/1/2020 | 11/30/2022 | £200.00 | | £0.00 | 0.00% |
| AB-SEA-PROD-MERAKI-MANAG... | RG-SEA-PROD-MERAKI-MA... | Monthly | 12/1/2020 | 11/30/2022 | £100.00 | | £11.34 | 11.34% |
| AB-SEA-PROD-MERAKI | RG-SEA-PROD-MERAKI (Res... | Monthly | 12/1/2020 | 11/30/2022 | £100.00 | | £0.00 | 0.00% |
| AB-UKS-CHECKPOINT-CLOUDG... | RG-UKS-CHECKPOINT-CLOU... | Monthly | 12/1/2020 | 11/30/2022 | £100.00 | | <£0.01 | 0.00% |
| AB-UKS-CHECKPOINT-CLOUDG... | AZ-UKS-CHECKPOINT-CLOU... | Monthly | 12/1/2020 | 11/30/2022 | £100.00 | | £0.00 | 0.00% |
| AB-UKS-CHECKPOINT-CLOUDG... | RG-UKS-CHECKPOINT-CLOU... | Monthly | 12/1/2020 | 11/30/2022 | £100.00 | | £0.00 | 0.00% |
| AB-UKS-GIS-WEB | RG-UKS-GIS-WEB (Resource ... | Monthly | 1/1/2021 | 12/31/2022 | £200.00 | | £0.00 | 0.00% |
| AB-UKS-STATIC-WEB-APP | RG-UKS-STATIC-WEB-APP (R... | Monthly | 1/1/2021 | 12/31/2022 | £100.00 | | £0.00 | 0.00% |
| AB-UKS-DEV-WVD-FULL | RG-UKS-DEV-WVD-FULL (Re... | Monthly | 2/1/2021 | 1/31/2023 | £100.00 | | £0.00 | 0.00% |
| AB-UKS-AZURE-DIGITAL-TWINS | RG-UKS-AZURE-DIGITAL-T... | Monthly | 2/1/2021 | 1/31/2023 | £100.00 | | £0.00 | 0.00% |
| AB-GMG-STORAGE | 03e8c261-92f4-4a4c-a6ce-a... | Monthly | 2/1/2021 | 1/31/2023 | £10,000.00 | | £7,363 | 73.63% |
| AB-UKS-PROD_WVD-IT | RG-UKS-PROD-WVD-IT (Res... | Monthly | 2/1/2021 | 1/31/2023 | £300.00 | | <£0.01 | 0.00% |
| AB-UKS-DEV-O365 | RG-UKS-DEV-O365 (Resourc... | Monthly | 3/1/2021 | 2/28/2023 | £100.00 | | £2.75 | 2.75% |
| RG-UKS-WATCHGUARD-FIREBO... | RG-UKS-WATCHGUARD-FIR... | Monthly | 3/1/2021 | 2/28/2023 | £100.00 | | £261.87 | 100.00% |
| AB-UKS-WATCHGUARD-FIREBO... | RG-UKS-WATCHGUARD-FIR... | Monthly | 3/1/2021 | 2/28/2023 | £100.00 | | <£0.01 | 0.00% |
| AB-UKS-IFS-P6 | RG-UKS-IFS-P6 (Resource gr... | Monthly | 3/1/2021 | 2/28/2023 | £100.00 | | £135.11 | 100.00% |
| AB-UKS-PURVIEW | RG-UKS-PURVIEW (Resource... | Monthly | 4/1/2021 | 3/31/2023 | £100.00 | | £0.00 | 0.00% |
| AB-UKS-WVD-IT | RG-UKS-WVD-IT (Resource ... | Monthly | 4/1/2021 | 3/31/2023 | £200.00 | | £41.92 | 20.96% |
| AB-UKS-WVD-OCEANIQ | RG-UKS-WVD-OceanIQ (Res... | Monthly | 4/1/2021 | 3/31/2023 | £200.00 | | £1,304 | 100.00% |

### 2.8.2 Recommendations

| Cost Management Recommendations | |
|---|---|
| Recommendation | Rationale |
| Consider the use of 1 year/3 year VM Reserved Instances for predictable, long-running workloads | Reduces cost by up to 72% in comparison to pay-as-you-go prices |
| Consider modernising application workloads to make use of PaaS services as a replacement of IaaS services | PaaS Services are cost-efficient and offer built-in reliability, scalability and security. For example, the replacement of SQL Server on VM with Azure SQL database. |
| Implement Azure Hybrid Use Benefit on SQL Servers using Azure Policy | Reduces costs of SQL Server licenses. |


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| | |
|---|---|
| (dependent on eligibility with Software Assurance) | |
| Implement shutdown schedules for development workloads using Azure Automation runbooks | Reduces overall compute costs. |
| Review and delete resources no longer in use | Reduces overall cloud spend; improves management and visibility of resources. |
| Right-size underused resources based on Azure Advisor recommendations | Reduces overall compute costs. |

## 2.9 Azure Advisor

### 2.9.1 Current State

The following attachment contains a full list of Azure Advisor recommendations exported as a CSV file:

Advisor Recommendations Re

The following attachment contains a full list of Azure Advisor recommendations exported as a PDF file:

Advisor Recommendations Re

### 2.9.2 Recommendations

| **Azure Advisor Recommendations** | |
|---|---|
| Recommendation | Rationale |
| Review Advisor recommendations with priority to high business impact recommendations | Improve the cost effectiveness, performance, Reliability and security of Azure resources. |
| Create Advisor alerts for high impact for cost, performance, reliability and operational excellence categories | Alerts will provide notifications for Advisor recommendations to allow issues to be identified and addressed proactively. |


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# 3 Identity and Access Management

Many consider identity to be the primary perimeter for security. This is a shift from the traditional focus on network security. Network perimeters keep getting more porous, and that perimeter defence cannot be as effective as it was before the explosion of BYOD devices and cloud applications. Azure Active Directory is the Azure solution for identity and access management. Azure AD is a multitenant, cloud-based directory and identity management service from Microsoft. It combines core directory services, application access management, and identity protection into a single solution.

A well-defined modern Identity strategy should allow organisations to protect their applications and data at the front gate with Azure identity and access management solutions, to defend against malicious login attempts and to safeguard credentials with risk-based access controls, identity protection tools, and strong authentication options – without disrupting productivity.

## 3.1 Azure AD Tenant Overview

A single Azure AD tenant has been created for the organisation.

| Azure AD Tenants | | | |
|---|---|---|---|
| Tenant Name | Tenant ID | Primary Domain | License |
| Global Marine Systems Limited | 2f411c57-c7ea-4a39-a409-dcf55c228745 | globalmarinesystems.onmicrosoft.com | Azure AD Premium P2 |

## 3.2 Azure AD Identity Synchronisation

### 3.2.1 Current State

A single Azure AD Connect server has been deployed to synchronise identities from Active Directory to Azure Active Directory.

| Azure AD Connect Servers | | | | |
|---|---|---|---|---|
| Server FQDN | Resource Group | Location | VM SKU | AAD Connect Version |
| GMSAZ019106.uk.gms.com | core-directoryservices | UK South | Standard_D2_v3 | 1.4.18.0 |

| Azure AD Connect Health Settings | |
|---|---|
| Setting | State |
| Use auto update to automatically update your installed Azure Active Directory Connect Health Agent with the latest version | On |
| Allow Microsoft access to your tenant's health data for troubleshooting purposes. This is typically done when | Off |


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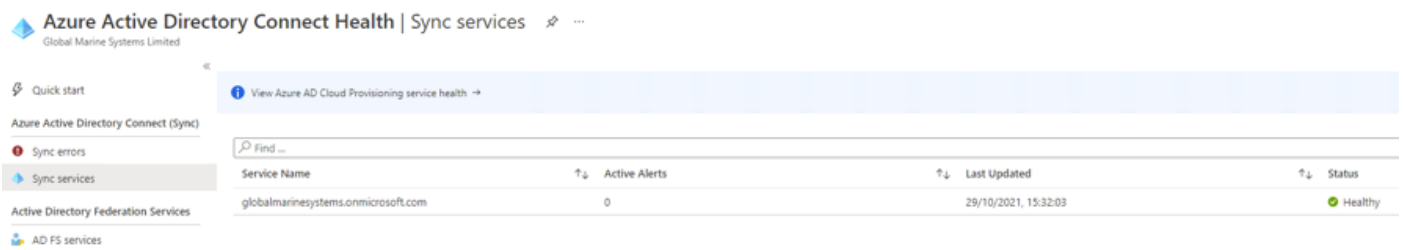
| | |
|---|---|
| you have an ongoing support call or if you are part of the preview program for this service. | |
| Use the notification to get notified when there are new alerts | On:<br>- Notify all global administrators<br>- No additional email recipients specified |

Pass-through Authentication (PTA) has been configured to allow users to sign in to both on-premises and cloud-based applications using the same passwords.

| Azure AD Connect PTA Agents | | | |
|---|---|---|---|
| Server FQDN | Resource Group | Location | VM SKU |
| GMSAZ019106.uk.gms.com | core-directoryservices | UK South | Standard_D2_v3 |
| GMSAZ019105.uk.gms.com | core-directoryservices | UK South | Standard_D2_v3 |

The state of AAD Connect is Healthy with zero active alerts:




There are currently twelve sync errors due to duplicate attributes. The following attachment contains a list of all sync errors:

AAD Connect Health Sync Errors.csv

### 3.2.2 Recommendations

| Azure AD Identity Synchronisation Recommendations | |
|---|---|
| Recommendation | Rationale |
| Deploy additional PTA agents in UK West | Provides regional resiliency for hybrid authentication in the event of an issue in UK South |
| Deploy an additional AAD Connect server in UK West in staging mode | Enables high availability for AAD Connect and allows new configuration changes to be tested in a staged approach |
| Review and resolve sync errors | Resolution of sync errors ensures that the synchronisation of identities between Active Directory and Azure Active Directory is up to date with no conflicting identities |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| | |
|---|---|
| Upgrade to the latest version of Azure AD Connect v2.0 | On 31 August 2022, all 1.x versions of Azure Active Directory (Azure AD) Connect will be retired because they include SQL Server 2012 components that will no longer be supported. |

## 3.3 Custom Domains

### 3.3.1 Current State

| Custom Domains | |
|---|---|
| Domain Name | Status |
| cwind.global | Verified |
| cwindtw.global | Verified |
| globalmarine.asia | Verified |
| globalmarine.co.uk | Verified |
| globalmarine.group | Verified |
| globalmarinesystems.com | Verified |
| globalmarinesystems.onmicrosoft.com | Available |
| globalmarinetest.co.uk | Verified |
| globaloffshore.co.uk | Verified |
| innovator.globalmarine.co.uk | Verified |
| normandclipper.globaloffshore.co.uk | Verified |
| oceaniq.co.uk | Verified |
| recorder.globalmarine.co.uk | Verified |
| recruitment.globalmarine.group | Verified |
| retriever.globalmarine.co.uk | Verified |
| sentinel.globalmarine.co.uk | Verified |
| sovereign.globalmarine.co.uk | Verified |
| symphony.globaloffshore.co.uk | Verified |
| wavemailuk.com | Verified |
| xvntn4j624p4.AEUWA03.msteams.8x8.com | Verified |
| Xvntn4j624p4.AUSEA02.msteams.8x8.com | Verified |

### 3.3.2 Recommendations

| Custom Domain Recommendations |
|---|
| No recommendations, all domains have been successfully verified. |


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## 3.4 Role Based Access Control (RBAC)

### 3.4.1 Current State

The following attachment contains a list of Azure AD Global Administrators:

Azure AD Global Administrators.csv

The following attachment contains a list of Azure Subscription role assignments:

Azure Subscription Role Assignments.csv

### 3.4.2 Recommendations

| RBAC Recommendations | |
|---|---|
| Recommendation | Rationale |
| Adopt least privileged access or Just Enough Access principles. Segregate duties and grant only the amount of access to users that they need to perform their jobs | • PIM best practices<br>• Instead of giving everybody unrestricted permissions in the Azure subscription or resources, allow only certain actions at a particular scope |
| The most appropriate Azure built-in RBAC role will be used as a first principle | • Management simplicity<br>• Microsoft do frequently revise and add/modify the existing list of built-in roles |
| Avoid resource-specific RBAC permissions, use management groups for enterprise-wide access control and resource groups for access control within subscriptions | • Complex and difficult to manage |
| Avoid user-specific RBAC, assign RBAC roles to groups in Azure AD. Use Azure-AD-only groups for Azure control-plane resources when granting access to resources. Add on-premises groups to the Azure-AD-only groups if needed | • Direct user assignments circumvent centralized management, greatly increasing the management required to prevent unauthorized access to restricted data |
| Create two emergency access accounts. These accounts should be cloud-only accounts that use the *.onmicrosoft.com domain and that are not federated or synchronized from an on-premises environment | • Mitigate the impact of accidental lack of administrative access<br>• AAD federation is unavailable<br>• Azure AD MFA service is unavailable<br>• Mobile phone or other networks might be unavailable |


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| | |
|---|---|
| Exclude at least one emergency account from phone-based multi-factor authentication, Conditional Access policies & Azure AD PIM | Emergency access in the event Azure AD MFA, Conditional Access, Azure AD PIM services are unavailable |
| Restrict emergency account use to only the times when it is necessary | Emergency access accounts are highly privileged, and they are not assigned to specific individuals |
| Validate emergency accounts regularly | Validation should be performed at regular intervals and for key changes |
| Leverage the benefits of PIM and integrate it as part of an effective Security and Identity baseline to be used in the Azure RBAC resource access workflow | Provide time-based and approval-based role activation to mitigate the risks of excessive, unnecessary, or misused access permissions on resources |

## 3.5 Multi-Factor Authentication (MFA)

### 3.5.1 Current State

| **Multi-Factor Authentication (MFA)** |
|---|
| Duo is a third-party service which is used to provide MFA |

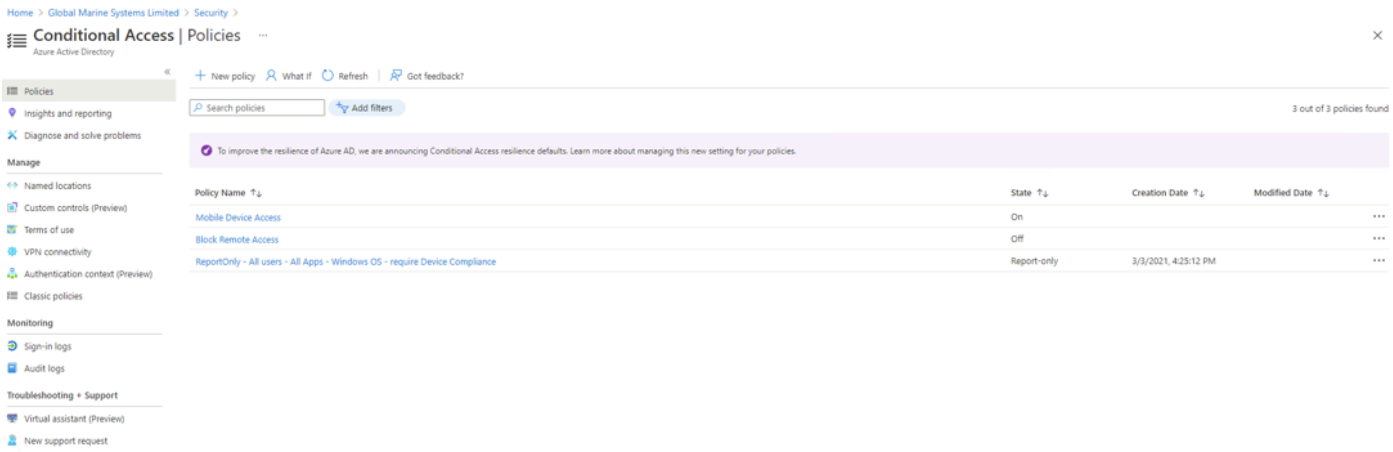
### 3.5.2 Recommendations

| **Multi-Factor Authentication (MFA)** |
|---|
| Azure AD is currently licensed with P2 licenses which includes Azure MFA. Evaluate the features and capabilities between Duo MFA and Azure MFA to determine the most appropriate solution to fulfil business requirements. |

## 3.6 Conditional Access Policies

### 3.6.1 Current State

The following conditional access policies have been created:





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Note: The Mobile Device Access policy is the only policy that is On, however, this appears to be for a proof of concept targeting a specific Azure AD group named Intune POC.

### 3.6.2 Recommendations

| Conditional Access Policies Recommendations | |
|---|---|
| Recommendation | Rationale |
| Integrate Duo with Conditional Access Policies to add MFA to Azure AD logons. https://duo.com/docs/azure-ca | Conditional Access Policies will allow for easier management of Duo MFA enablement to secure Azure AD applications. |
| Implement additional Conditional Access Policies to secure Azure AD applications. Common examples include:<br>- Requiring MFA for users with administrative roles<br>- Requiring MFA for Azure management tasks<br>- Blocking sign-ins for users attempting to use legacy authentication protocols<br>- Blocking or granting access from specific locations<br>- Blocking risky sign-in behaviours<br>- Requiring organization-managed devices for specific applications | Improves identity security posture in adherence to zero-trust security and enforces organisational policies. |
| Use report-only mode to evaluate the impact of Conditional Access policies before enabling them in their environment. | Allows administrators to evaluate the impact of Conditional Access policies before enabling them in their environment |

## 3.7 Azure AD Administrative Units

### 3.7.1 Current State

| Azure AD Administrative Units |
|---|
| Administrative Units are not in use |

### 3.7.2 Recommendations

| Azure AD Administrative Units Recommendations | |
|---|---|
| Recommendation | Rationale |
| Consider implementing Administrative Units to restrict administrative scope | An administrative unit is an Azure AD resource that can be a container for other Azure AD resources. An administrative unit can contain only users and groups. |


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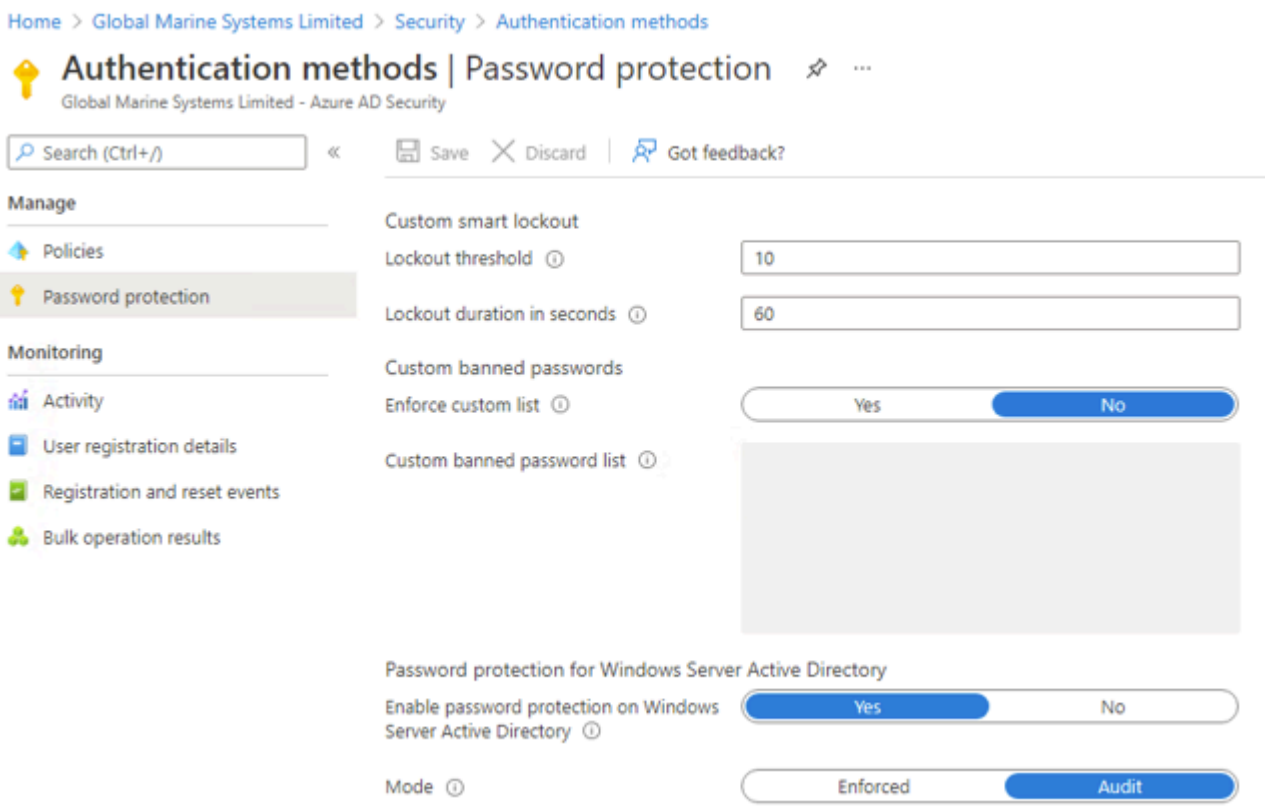
| | |
|---|---|
| | Administrative units restrict permissions in a role to any portion of your organization that you define. For example, administrative units could be used to delegate the Helpdesk Administrator role to regional support specialists, so they can manage users only in the region that they support. |

## 3.8 Azure AD Password Protection

### 3.8.1 Current State

| Azure AD Password Protection |
|---|
| Azure AD Password Protection has been enabled in Audit mode |




### 3.8.2 Recommendations

| Azure AD Password Protection Recommendations | |
|---|---|
| Recommendation | Rationale |
| Carry out testing in Audit mode by attempting to reset a test user account password with a simple password. Launch Event Viewer on the server | Testing in Audit Mode will validate the setup of DC agents and connectivity to Azure AD. |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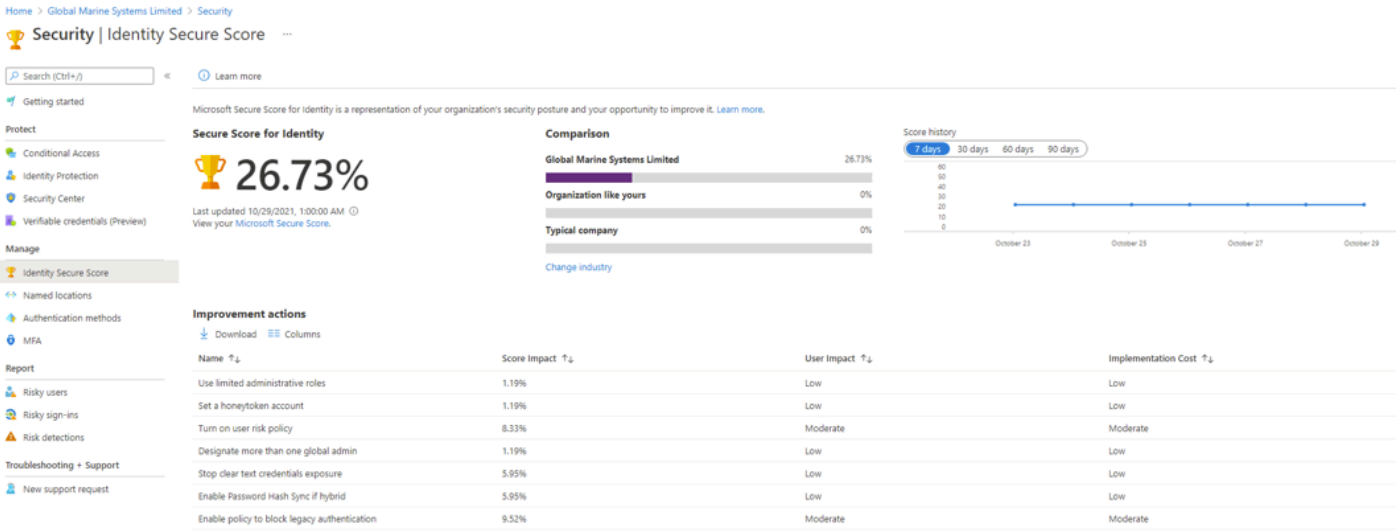
| | |
|---|---|
| hosting the DC agent and navigate to Applications and Services Logs -> Microsoft -> AzureADPasswordProtection -> DCAgent -> Admin and check events are being logged in relation to the failed password reset (event ID 10016, 30004, 10017, 30005). | |
| Once testing has been successfully completed, change the mode from Audit to Enforced. | Enforcing password protection will provide active protection against common passwords. |

## 3.9 Azure AD Secure Score

### 3.9.1 Current State

The current Secure Score for Identity is 26.73%.




The following attachment contains a list of improvement actions:

AAD Secure Score for Identity Improvement

### 3.9.2 Recommendations

| Azure AD Secure Score Recommendations | |
|---|---|
| Recommendation | Rationale |
| Review Azure AD Secure Score improvement actions with priority to high score impact actions | Improves identity security posture |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# 4 Networking

## 4.1 Overview

| Network Overview |
| --- |
| WatchGuard Firewall is used to provide ingress and egress to all Azure servers and services. |
| A Cisco Meraki device (AZUKS01-FW1) is deployed in Azure as a VPN concentrator to provide seamless integration with on-premise traffic and site-to-site VPNs. |
| Azure public load balancers are deployed to distribute traffic between healthy virtual machine instances and also provide high-availability and resilience to specific services such as Exchange Online, DUO MFA and Backup services. |
| Network Security Groups have been created and associated to each subnet to secure traffic between subnets within the same virtual network. |
| Route tables have been created and associated to each subnet with a default route to WatchGuard Firewall. |

## 4.2 Virtual Networks

| Virtual Networks | | | |
| --- | --- | --- | --- |
| Name | Resource Group | Address Space | Location |
| Core-AZ01-UKS | core-networking | 10.15.0.0/16 | UK South |
| Core-AZ01-UKS-asr | core-networking-asr | 10.15.0.0/16 | UK West |
| Core-AZ02-UKW | core-networking | 10.16.0.0/16 | UK West |
| Core-AZ03-SEA | rg-sea-prod-singapore | 10.17.0.0/16 | Southeast Asia |
| RG-UKS-WVD-OceanIQ-vnet | rg-uks-wvd-oceaniq | 10.0.0.0/16 | UK South |
| TestVnet | core-networking | 10.1.0.0/28 | UK West |
| WatchguardDevVnet | rg-uks-watchguard-dev | 10.20.20.0/24 | UK South |
| WatchguardVnet | rg-uks-watchguard-firebox-cloud | 10.10.10.0/24 | UK South |

## 4.3 Load Balancers

| Load Balancers | | | |
| --- | --- | --- | --- |
| Name | Resource Group | Location | Load Balancer Type |
| gms-az01-ex-lb | services-exchange | UK South | Public |
| GMS-Backup-SS1-Mirror-SEA | rg-sea-prod-singapore | Southeast Asia | Public |
| GMS-Backup-SS1-UKS | core-networking | UK South | Public |
| GMS-Backup-SS3-Mirror-UKS | core-networking | UK South | Public |
| GMS-Backup-SS3-UKW | core-networking | UK West | Public |
| GMS-UKS-DUO-LB | core-networking | UK South | Public |
| LB-UKS-IFS-P6-EXT | rg-uks-ifs-p6 | UK South | Public |


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## 4.4 Network Security Groups

| Network Security Groups | | | |
|---|---|---|---|
| Name | Resource Group | Location | Associations |
| AZ01-AD-Servers | core-networking | UK South | 2 subnets, 4 network interfaces |
| AZ01-Checkpoint-Test | core-networking | UK South | 0 subnets, 0 network interfaces |
| AZ01-DMZ | core-networking | UK South | 2 subnets, 0 network interfaces |
| AZ01-Email-Servers | core-networking | UK South | 1 subnets, 0 network interfaces |
| az01-exchange | services-exchange | UK South | 1 subnets, 0 network interfaces |
| AZ01-Mgmt-Servers | core-networking | UK South | 1 subnets, 0 network interfaces |
| AZ01-UKS-DMS | core-networking | UK South | 1 subnets, 0 network interfaces |
| AZ01-UKS-WVD | core-networking | UK South | 0 subnets, 0 network interfaces |
| AZ02-AD-Servers | core-networking | UK West | 1 subnets, 0 network interfaces |
| AZ02-DMZ | core-networking | UK West | 1 subnets, 0 network interfaces |
| AZ02-Email-Servers | core-networking | UK West | 1 subnets, 0 network interfaces |
| AZ02-File-Servers | core-networking | UK West | 0 subnets, 0 network interfaces |
| AZ02-Mgmt-Servers | core-networking | UK West | 1 subnets, 0 network interfaces |
| AZ03-AD-Servers | rg-sea-prod-singapore | Southeast Asia | 1 subnets, 0 network interfaces |
| AZ03-AD-Servers-Test | rg-sea-prod-singapore | Southeast Asia | 0 subnets, 0 network interfaces |
| AZ03-DMZ | rg-sea-prod-singapore | Southeast Asia | 1 subnets, 0 network interfaces |
| AZ03-Mgmt-Servers | rg-sea-prod-singapore | Southeast Asia | 1 subnets, 0 network interfaces |
| AZUKSSQLMGMT01-NSG | core-networking | UK West | 0 subnets, 0 network interfaces |
| basicNsgWatchGuardVMEth1-NSG | rg-uks-watchguard-firebox-cloud | UK South | 0 subnets, 0 network interfaces |
| gms-az-mgmt01-nsg | core-management | UK South | 0 subnets, 0 network interfaces |


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| | | | |
|---|---|---|---|
| JumpHost-nsg | core-networking | UK West | 0 subnets, 0 network interfaces |
| JumpHostfromWatchguardVnet-nsg | rg-uks-watchguard-firebox-cloud | UK South | 0 subnets, 0 network interfaces |
| JumpHostVM-nsg | core-networking | UK West | 0 subnets, 0 network interfaces |
| nsg-azuksgbsql2 | core-networking | UK South | 1 subnets, 0 network interfaces |
| NSG-WatchGuardDevEth0None | rg-uks-watchguard-dev | UK South | 0 subnets, 0 network interfaces |
| NSG-WatchGuardVMEth0Allow | rg-uks-watchguard-firebox-cloud | UK South | 0 subnets, 1 network interfaces |
| TestVMfromWatchguardVnet-nsg | rg-uks-watchguard-firebox-cloud | UK South | 0 subnets, 0 network interfaces |
| UKS-Image-OceanIQ-nsg | rg-uks-wvd-oceaniq | UK South | 0 subnets, 1 network interfaces |
| AZ01-AD-Servers | core-networking | UK South | 2 subnets, 4 network interfaces |

## 4.5 Route Tables

| Route Tables | | | |
|---|---|---|---|
| Name | Resource Group | Location | Associations |
| Meraki-SEA-RT01 | rg-sea-prod-singapore | Southeast Asia | 3 subnet associations |
| Meraki-SEA-RT01-Test | rg-sea-prod-singapore | Southeast Asia | 0 subnet associations |
| Meraki-UKS-RT01 | meraki-uks | UK South | 7 subnet associations |
| Meraki-UKW-RT01 | meraki-uks | UK West | 2 subnet associations |
| rt-azuksgbsql2 | core-networking | UK South | 1 subnet associations |
| UDRTOWATCHGAURDVM | core-networking | UK West | 1 subnet associations |
| WatchGuardDev-External-routes | rg-uks-watchguard-dev | UK South | 1 subnet associations |
| WatchGuardDev-Trusted-routes | rg-uks-watchguard-dev | UK South | 1 subnet associations |
| WatchGuardVM-External-routes | rg-uks-watchguard-firebox-cloud | UK South | 1 subnet associations |
| WatchGuardVM-Trusted-routes | rg-uks-watchguard-firebox-cloud | UK South | 1 subnet associations |


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## 4.6 Public IP Addresses

| Public IP Addresses | | | |
|---|---|---|---|
| Name | Resource Group | IP Address | Location |
| AZ01-ERG | core-networking | 51.104.199.62 | UK South |
| AZUKSIFSGW01-ip | rg-uks-ifs-p6 | 20.77.107.235 | UK South |
| cj5aue3mqjasmIP | rg-sea-prod-meraki-managed | 40.119.204.99 | Southeast Asia |
| GatewayIP-z6wt6w2mrnids | core-networking | - | UK South |
| GBAZ-DUO-2-ip | core-networking | 40.81.145.85 | UK South |
| GBAZ-MGMT-01-ip | core-networking | 51.105.59.44 | UK South |
| GBAZ-WEB-01-ip | core-networking | 51.141.239.193 | UK South |
| GBAZDUO2ip902 | core-networking | 40.81.147.12 | UK South |
| gms-az01-ex-lb-ip | services-exchange | 51.145.124.177 | UK South |
| GMS-Backup-SS1-Mirror-SEA | rg-sea-prod-singapore | - | Southeast Asia |

## 4.7 Recommendations

| Network Recommendations |
|---|
| Azure Firewall is a cloud-native Azure service offering built-in high availability, scalability and IDS/IPS. Evaluate the features and capabilities of Azure Firewall in comparison with WatchGuard to determine the most appropriate solution to fulfil the requirements. |
| Azure VPN Gateway is a cloud-native Azure service offering built-in high availability, scalability and resiliency. Evaluate the features and capabilities of Azure VPN Gateway in comparison with Cisco Meraki to determine the most appropriate solution to fulfil the requirements. |
| Consider implementing DDoS Protection Standard to mitigate against DDoS threats for public facing workloads. |
| Consider implementing Azure Front Door or Application Gateway with Web Application Firewall to protect web applications from common security threats. |
| Prevent direct assignment of public IP address to virtual machines, with the exception of Network Virtual Appliances (NVAs). |
| Prevent IP forwarding on virtual machine network interfaces, with the exception of Network Virtual Appliances (NVAs). |
| Review and delete unassociated Network Security Groups |
| Review and delete unassociated Public IP Addresses |
| Review and delete unassociated Route Tables |
| Use Private Endpoints where possible to secure access to PaaS resources |


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# 5 Compute

## 5.1 Virtual Machines

### 5.1.1 Current State

| Virtual Machines | | | | | | |
|---|---|---|---|---|---|---|
| Name | Resource Group | Location | Status | Operating System | Size | Public IP |
| AVD-IMAGE-OIQ | rg-uks-wvd-oceaniq | UK South | Stopped (deallocated) | Windows | Standard_D2s_v3 | - |
| AZ-UKS-Migrate | rg-uks-azure-migrate | UK South | Stopped (deallocated) | Windows | Standard_D8s_v3 | - |
| AZSEASINDOC01 | rg-sea-prod-singapore | Southeast Asia | Running | Windows | Standard_D2s_v3 | - |
| AZSEASINDOC02 | rg-sea-prod-singapore | Southeast Asia | Running | Windows | Standard_D2s_v3 | - |
| AZSEASINFS01 | rg-sea-prod-singapore | Southeast Asia | Running | Windows | Standard_DS3_v2 | - |
| AZSEASINFW01 | rg-sea-prod-meraki-managed | Southeast Asia | Running | Linux | Standard_F4s_v2 | 40.119.204.99 |
| AZUKS01-FW1 | meraki-uksemtu7tifmoczk | UK South | Running | Linux | Standard_D2_v2 | 51.140.95.220 |
| AZUKSDOC01 | core-directoryservices | UK South | Running | Windows | Standard_D2s_v3 | - |
| AZUKSDOC02 | core-directoryservices | UK South | Running | Windows | Standard_D2s_v3 | - |
| AZUKSDUO01 | core-networking | UK South | Running | Windows | Standard_DS1_v2 | - |
| AZUKSIFSGW01 | rg-uks-ifs-p6 | UK South | Running | Windows | Standard_D4s_v3 | 20.77.107.235 |
| AZUKSNPS01 | core-networking | UK South | Running | Windows | Standard_DS1_v2 | - |
| AZUKSRDPOIQ1 | rg-uks-wvd-oceaniq | UK South | Stopped (deallocated) | Windows | Standard_D4s_v3 | - |
| AZUKSWVDITS-0 | rg-uks-wvd-it | UK South | Stopped (deallocated) | Windows | Standard_D4s_v3 | - |
| AZUKSWVDOIQ-0 | rg-uks-wvd-oceaniq | UK South | Stopped (deallocated) | Windows | Standard_D4s_v3 | - |
| GBAZ-DFIRT-01 | rg-uks-azure-migrate-sql | UK South | Stopped (deallocated) | Windows | Standard_D2s_v3 | - |
| GBAZ-DUO-2 | core-networking | UK South | Running | Windows | Standard_B2s | 40.81.147.12 |
| GBAZ-FS-02 | core-networking | UK South | Running | Windows | Standard_DS3_v2 | - |
| GBAZ-FS-03 | core-networking | UK South | Running | Windows | Standard_DS3_v2 | - |
| GBAZ-MGMT-01 | core-networking | UK South | Running | Windows | Standard_B4ms | 51.105.59.44 |
| GBAZ-SQL-OIQ-01 | rg-uks-azure-sql-mi | UK South | Running | Windows | Standard_D2s_v3 | - |
| GBAZ-WEB-01 | core-networking | UK South | Running | Windows | Standard_DS2_v2 | 51.141.239.193 |
| gms-az01-ex01 | services-exchange | UK South | Running | Windows | Standard_D4s_v3 | - |
| GMSAZ019105 | core-directoryservices | UK South | Running | Windows | Standard_D2_v3 | - |
| GMSAZ019106 | core-directoryservices | UK South | Running | Windows | Standard_D2_v3 | - |


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| | | | | | | |
|---|---|---|---|---|---|---|
| IT-TEST-WVD-0 | rg-uks-wvd-it | UK South | Stopped (deallocated) | Windows | Standard_D2s_v3 | - |
| OIQPOCImage | rg-uks-wvd-oceaniq | UK South | Stopped (deallocated) | Windows | Standard_D2s_v3 | - |
| OIQRestore | rg-uks-wvd-it | UK South | Stopped (deallocated) | Windows | Standard_NV6 | - |
| TestVM | core-networking | UK West | Stopped (deallocated) | Windows | Standard_D2s_v3 | - |
| uk-avdpoc | rg-uks-wvd-oceaniq | UK South | Running | Windows | Standard_NV6 | - |
| WatchGuardDev | rg-uks-watchguard-dev | UK South | Running | Linux | Standard_A1_v2 | 20.90.97.24 |
| WatchGuardVM | rg-uks-watchguard-firebox-cloud | UK South | Running | Linux | Standard_A8_v2 | 20.108.21.51 |
| WVD-GOLDEN-IMAGE | rg-uks-wvd-it | UK South | Stopped (deallocated) | Windows | Standard_D2s_v3 | - |

### 5.1.2 Recommendations

| **Virtual Machine Recommendations** | |
|---|---|
| Recommendation | Rationale |
| Review and delete stopped and deallocated machines | Reduces compute and storage costs |
| Review the usage of public IP addresses | Use public IP address only where necessary, for example, Watchguard firewall VM |
| Review the usage of A-series and B-series VMs | A-series and B-series VMs are intended for use in development environments only |
| Consider modernising application workloads to make use of PaaS services as a replacement of IaaS services | PaaS Services are cost-efficient and offer built-in reliability, scalability and security. For example, the replacement of SQL Server on VM with Azure SQL database. |
| Ensure VMs are named in accordance to naming standards | Consistent naming of resources is crucial to identifying resources not only in the Azure portal and in bills, but also in scripts that support Azure DevOps automation/orchestration for release management. |


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# 6 Storage

## 6.1 Managed Disks

### 6.1.1 Current State

| Managed Disks | | | | | |
|---|---|---|---|---|---|
| Name | Storage Account Type | Size (GB) | Owner | Resource Group | Disk State |
| AVD-IMAGE-OIQ_OsDisk_1_a314abf170cc4d42a131a0b3b9ea11ed | Standard SSD LRS | 127 | AVD-IMAGE-OIQ | rg-uks-wvd-oceaniq | Reserved |
| AZ-UKS-Migrate_OsDisk_1_5247fb3719d94ea8af89aef6b3c718bf | Standard HDD LRS | 127 | - | rg-uks-azure-migrate | Unattached |
| az-uks-poc-0_datadisk_1 | Standard SSD LRS | 128 | - | rg-uks-wvd-oceaniq | Unattached |
| AZ-UKS-POC-0_OsDisk_1_735ff6ff19ac4490b9db026cd937ca33 | Standard SSD LRS | 127 | - | rg-uks-wvd-oceaniq | Unattached |
| azseainfs01_data2 | Standard SSD LRS | 1024 | AZSEASINFS01 | rg-sea-prod-singapore | Attached |
| AZSEASINDOC01_OsDisk_1_cf412112083242618634de588b1dbd52 | Standard SSD LRS | 127 | AZSEASINDOC01 | rg-sea-prod-singapore | Attached |
| AZSEASINDOC02_OsDisk_1_7803884f6ef541ecbffe185d125be286 | Standard SSD LRS | 127 | AZSEASINDOC02 | rg-sea-prod-singapore | Attached |
| azseasinfs01_data | Standard SSD LRS | 800 | AZSEASINFS01 | rg-sea-prod-singapore | Attached |
| AZSEASINFS01_OsDisk_1_bf274ef46b104206bc5a360077225be0 | Standard SSD LRS | 127 | AZSEASINFS01 | rg-sea-prod-singapore | Attached |
| AZSEASINFW01_OsDisk_1_731281e77bea4063892a4a205510006c | Premium SSD LRS | 2 | AZSEASINFW01 | rg-sea-prod-meraki-managed | Attached |
| AZUKS01-FW1_OsDisk_1_3d99ca1a1c3640b99ba66cef529b0575 | Standard HDD LRS | 2 | AZUKS01-FW1 | meraki-uksemtu7tifmoczk | Attached |
| AZUKSDOC01_OsDisk_1_9fc67995b8864e26abbf0827c5d292b5 | Standard HDD LRS | 127 | AZUKSDOC01 | core-directoryservices | Attached |
| AZUKSDOC02_OsDisk_1_a125126846634c5a913c1bb60c57678f | Standard HDD LRS | 127 | AZUKSDOC02 | core-directoryservices | Attached |
| AZUKSDUO01_disk1_e742e96139c64a5cada8cc49e80d2d31 | Standard SSD LRS | 127 | AZUKSDUO01 | core-networking | Attached |
| AZUKSIFSGW01_OsDisk_1_47efcac8d919421487265f58e378b148 | Standard HDD LRS | 127 | AZUKSIFSGW01 | rg-uks-ifs-p6 | Attached |
| azuksmigrate-osdisk-20210821-095247 | Standard HDD LRS | 127 | AZ-UKS-Migrate | rg-uks-azure-migrate | Reserved |


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| | | | | | |
|---|---|---|---|---|---|
| AZUKSNPS01_OsDisk_1_af4475a59b0f4540b5822eb0554cb9ee | Standard HDD LRS | 127 | AZUKSNPS01 | core-networking | Attached |
| azuksrdpoiq1-osdisk-20210506-142702 | Standard SSD LRS | 127 | AZUKSRDPOIQ1 | rg-uks-wvd-oceaniq | Reserved |
| AZUKSRDPOIQ1_OsDisk_1_69ecbc3cfbd94dea9473cd5ba4dd2e69 | Standard SSD LRS | 127 | - | rg-uks-wvd-oceaniq | Unattached |
| AZUKSWVDITS-0_disk1_3bf9133d3cbe4a02912c3153871e5402 | Standard SSD LRS | 127 | AZUKSWVDITS-0 | rg-uks-wvd-it | Reserved |
| AZUKSWVDOIQ-0_OsDisk_1_d5e9ab94b90245a4b65e602bb7b23a1f | Standard SSD LRS | 127 | AZUKSWVDOIQ-0 | rg-uks-wvd-oceaniq | Reserved |
| GBAZ-DFIRT-01_disk1_92351415ef224d7786d1624bc4c1c554 | Standard SSD LRS | 127 | GBAZ-DFIRT-01 | rg-uks-azure-migrate-sql | Reserved |
| GBAZ-DUO-2_OsDisk_1_3a85e9936a8b4e68a3057c6e7e662486 | Standard HDD LRS | 127 | GBAZ-DUO-2 | core-networking | Attached |
| GBAZ-FS-02_DataDisk_0 | Standard HDD LRS | 8192 | GBAZ-FS-02 | core-networking | Attached |
| GBAZ-FS-02_DataDisk_1 | Standard HDD LRS | 8192 | GBAZ-FS-02 | core-networking | Attached |
| GBAZ-FS-02_DataDisk_2 | Standard HDD LRS | 8192 | GBAZ-FS-02 | core-networking | Attached |
| GBAZ-FS-02_DataDisk_3 | Standard HDD LRS | 8192 | GBAZ-FS-02 | core-networking | Attached |
| GBAZ-FS-02_DataDisk_4 | Standard HDD LRS | 8192 | GBAZ-FS-02 | core-networking | Attached |
| GBAZ-FS-02_DataDisk_5 | Standard HDD LRS | 8192 | GBAZ-FS-02 | core-networking | Attached |
| GBAZ-FS-02_DataDisk_6 | Standard HDD LRS | 8192 | GBAZ-FS-02 | core-networking | Attached |
| GBAZ-FS-02_DataDisk_7 | Standard HDD LRS | 8192 | GBAZ-FS-02 | core-networking | Attached |
| GBAZ-FS-02_DataDisk_8 | Standard HDD LRS | 8192 | GBAZ-FS-02 | core-networking | Attached |
| GBAZ-FS-02_DataDisk_9 | Standard HDD LRS | 8192 | GBAZ-FS-02 | core-networking | Attached |
| GBAZ-FS-02_OsDisk_1_aec55bd4738f4931b09ee95a945ba873 | Standard SSD LRS | 127 | GBAZ-FS-02 | core-networking | Attached |
| gbaz-fs-03_datadisk0 | Standard HDD LRS | 11000 | GBAZ-FS-03 | core-networking | Attached |
| GBAZ-FS-03_OsDisk_1_2c181dc326d04a8c809f4bbf92c740db | Standard SSD LRS | 127 | GBAZ-FS-03 | core-networking | Attached |


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| | | | | | |
|---|---|---|---|---|---|
| GBAZ-MGMT-01_OsDisk_1_ed7e639f0d264e9b8fe24484c835d592 | Standard HDD LRS | 127 | GBAZ-MGMT-01 | core-networking | Attached |
| GBAZ-MGMT-01_OsDisk_1_ed7e639f0d264e9b8fe24484c835d592-ASRReplica | Standard HDD LRS | 127 | - | core-networking-asr | ActiveSAS |
| GBAZ-SQL-OIQ-01_DataDisk_0 | Premium SSD LRS | 1024 | - | rg-uks-azure-sql-mi | Unattached |
| GBAZ-SQL-OIQ-01_DataDisk_1 | Premium SSD LRS | 1024 | - | rg-uks-azure-sql-mi | Unattached |
| gbaz-sql-oiq-01_datadisk_2 | Premium SSD LRS | 1024 | GBAZ-SQL-OIQ-01 | rg-uks-azure-sql-mi | Attached |
| gbaz-sql-oiq-01_datadisk_3 | Premium SSD LRS | 1024 | GBAZ-SQL-OIQ-01 | rg-uks-azure-sql-mi | Attached |
| GBAZ-SQL-OIQ-01_OsDisk_1_544fd2c79e96499b95c781cfad4c6c3a | Premium SSD LRS | 127 | GBAZ-SQL-OIQ-01 | rg-uks-azure-sql-mi | Attached |
| GBAZ-WEB-01_DataDisk_0 | Standard SSD LRS | 1023 | GBAZ-WEB-01 | core-networking | Attached |
| GBAZ-WEB-01_OsDisk_1_3d11fff1df61470fa59d50798d2f2c64 | Standard HDD LRS | 127 | GBAZ-WEB-01 | core-networking | Attached |
| gms-az01-ex01_OsDisk_1_702c2f0f649d48efbcffeff5ab3d1c9a | Premium SSD LRS | 256 | - | services-exchange | Unattached |
| gms-az01-ex02_OsDisk_1_bd692f9b18834bfda2618103d397c1e3 | Premium SSD LRS | 120 | - | services-exchange | Unattached |
| gms-az01-ex03_OsDisk_1_834dd059bf9045cf9b113e580ca15075 | Premium SSD LRS | 127 | - | services-exchange | Unattached |
| GMSAZ019105_OsDisk_1_ff453ab9c6bf4d548dd561eb5831951a | Standard HDD LRS | 127 | GMSAZ019105 | core-directoryservices | Attached |
| GMSAZ019105_OsDisk_1_ff453ab9c6bf4d548dd561eb5831951a-ASRReplica | Standard HDD LRS | 127 | - | core-directoryservices-asr | ActiveSAS |
| GMSAZ019106_OsDisk_1_77e16ba110694133b4f82027cf7d15bc | Standard HDD LRS | 127 | GMSAZ019106 | core-directoryservices | Attached |
| GMSAZ019106_OsDisk_1_77e16ba110694133b4f82027cf7d15bc-ASRReplica | Standard HDD LRS | 127 | - | core-directoryservices-asr | ActiveSAS |
| gmsaz01ex01-osdisk-20210821-090210 | Premium SSD LRS | 256 | - | services-exchange | Unattached |
| gmsaz01ex01-osdisk-20210821-101517 | Premium SSD LRS | 256 | gms-az01-ex01 | services-exchange | Attached |
| IT-TEST-WVD-0_OsDisk_1_b0ea09662c7f4438b9a22a1f3bb46c7b | Standard SSD LRS | 127 | IT-TEST-WVD-0 | rg-uks-wvd-it | Reserved |
| JumpHost_OsDisk_1_65047f5e0b2d47e58887ddb00b4c3e5d | Standard SSD LRS | 127 | - | core-networking | Unattached |


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| | | | | | |
|---|---|---|---|---|---|
| JumpHost_OsDisk_1_aff2f326bc0d45b9bf2f92e7970ec6d0 | Standard HDD LRS | 127 | - | core-networking | Unattached |
| Jumphost_OsDisk_1_ec69b07fe24c4a3ab051c0b9b9a4e689 | Premium SSD LRS | 127 | - | core-networking | Unattached |
| JumpHostfromWatchguardVnet_disk1_90e8c638d95f4438b8cfb67f24176681 | Standard SSD LRS | 127 | - | rg-uks-watchguard-firebox-cloud | Unattached |
| JumpHostVM_disk1_cb8371d2bfad4f98aa13dd31b776c2b7 | Standard HDD LRS | 127 | - | core-networking | Unattached |
| OIQPOCImage-osdisk | Standard SSD LRS | 128 | OIQPOCImage | rg-uks-wvd-oceaniq | Reserved |
| oiqrestore-datadisk-000-20210730-130521 | Standard SSD LRS | 128 | OIQRestore | rg-uks-wvd-it | Reserved |
| oiqrestore-osdisk-20210730-130521 | Standard SSD LRS | 127 | OIQRestore | rg-uks-wvd-it | Reserved |
| TestVM_OsDisk_1_14c74dfa85a04bc1b200f427c30b9995 | Standard HDD LRS | 127 | TestVM | core-networking | Reserved |
| TestVMfromWatchguardVnet_disk1_ca91f941c81c4565be01a163d081a5b0 | Standard SSD LRS | 127 | - | rg-uks-watchguard-firebox-cloud | Unattached |
| uk-avdpoc-osdisk | Standard SSD LRS | 128 | uk-avdpoc | rg-uks-wvd-oceaniq | Attached |
| UKS-Image-OceanIQ_OsDisk_1_b8e23ebd3f184218a13a15368e0a07b0 | Standard SSD LRS | 127 | - | rg-uks-wvd-oceaniq | Unattached |
| WatchGuardDev | Standard HDD LRS | 4 | WatchGuardDev | rg-uks-watchguard-dev | Attached |
| WatchGuardVM | Standard HDD LRS | 4 | WatchGuardVM | rg-uks-watchguard-firebox-cloud | Attached |
| WVD-GOLDEN-IMAGE_OsDisk_1_64cc2139b7b942feaf236073c3900195 | Standard SSD LRS | 127 | WVD-GOLDEN-IMAGE | rg-uks-wvd-it | Reserved |

### 6.1.2 Recommendations

| Managed Disk Recommendations | |
|---|---|
| Recommendation | Rationale |
| Review and delete unattached disks | Reduces overall storage costs |
| Consider the modernisation of file servers to Azure Files | Azure Files offers fully managed SMB file shares and reduces overall infrastructure costs |

## 6.2 Storage Accounts

### 6.2.1 Current State


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| Storage Accounts | | | | |
|---|---|---|---|---|
| Name | Kind | SKU | Resource Group | Location |
| cdgcq6gms0backupasrcache | Storage | Standard_LRS | core-networking | UK South |
| chartingstorage1 | FileStorage | Premium_LRS | gm-azure-storage | UK South |
| corenetworkingdiag229 | Storage | Standard_LRS | core-networking | East US |
| corenetworkingdiag275 | Storage | Standard_LRS | core-networking | North Europe |
| corenetworkingdiag999 | Storage | Standard_LRS | core-networking | UK South |
| csa03e8c26192f4x4a4cxa6c | StorageV2 | Standard_LRS | cloud-shell-storage-northeurope | North Europe |
| csb03e8c26192f4x4a4cxa6c | StorageV2 | Standard_LRS | cloud-shell-storage-westeurope | West Europe |
| csb10032000cb1e6eb1 | StorageV2 | Standard_LRS | cloud-shell-storage-westeurope | West Europe |
| csb10032000cb23f916 | StorageV2 | Standard_LRS | cloud-shell-storage-westeurope | West Europe |
| csb1003200138ca6c58 | StorageV2 | Standard_LRS | cloud-shell-storage-westeurope | West Europe |
| csb100320015f69c96a | StorageV2 | Standard_LRS | cloud-shell-storage-westeurope | West Europe |
| cssa64457de4fb4b8e20100d | StorageV2 | Standard_LRS | rg-uks-wvd-oceaniq | UK South |
| dkfproject | StorageV2 | Standard_LRS | core-networking | West Europe |
| gbazfs01 | Storage | Standard_LRS | core-networking | West Europe |
| gmazurestoragedkf | Storage | Standard_LRS | gm-azure-storage | UK South |
| gmguksaberdeen | StorageV2 | Standard_LRS | rg-uks-prod-aberdeen-storage | UK South |
| gmsblythstoreuks | StorageV2 | Standard_LRS | gm-azure-storage | UK South |
| rgseaprodsingaporediag | Storage | Standard_LRS | rg-sea-prod-singapore | Southeast Asia |
| rgukscheckpointcloudg179 | StorageV2 | Standard_LRS | rg-uks-checkpoint-cloudguard | UK South |
| rguksdev0365datalakegen2 | StorageV2 | Standard_LRS | rg-uks-dev-o365 | UK South |
| rguksdevo365 | StorageV2 | Standard_LRS | rg-uks-dev-o365 | UK South |
| spsiteprovisioning202107 | Storage | Standard_LRS | o365_resources | East US |
| sqlvabcoi3iwxde5oa | StorageV2 | Standard_LRS | rg-uks-azure-sql-mi | UK South |
| sqlvas01 | Storage | Standard_LRS | rg-uks-azure-sql-mi | UK South |
| stgazukswvdoiq07f712322 | Storage | Standard_GRS | rg-uks-wvd-oceaniq | UK South |
| stgittestwvd0f7104674 | Storage | Standard_GRS | rg-uks-wvd-it | UK South |
| stguksavdpoc0c3832193 | Storage | Standard_GRS | rg-uks-wvd-oceaniq | UK South |
| stn80c5ba94d38b31cd8b46d | Storage | Standard_LRS | rg-uks-wvd-oceaniq | UK South |
| storageaccounto365ra6e6 | Storage | Standard_LRS | o365_resources | UK South |
| storageaccounto365raa50 | Storage | Standard_LRS | o365_resources | Central US |
| stseafileshare | FileStorage | Premium_ZRS | rg-sea-prod-singapore | Southeast Asia |
| stuksazuremigratesql | StorageV2 | Standard_GRS | rg-uks-azure-migrate-sql | UK South |
| stuksdevfileshare | StorageV2 | Standard_ZRS | rg-uks-dev-test | UK South |
| stuksfileshare1 | FileStorage | Premium_ZRS | gm-azure-storage | UK South |
| stuksifsp6diag | StorageV2 | Standard_RAGRS | rg-uks-ifs-p6 | UK South |
| stuksoceanwvd | FileStorage | Premium_LRS | rg-uks-wvd-oceaniq | UK South |
| stuksprodrovcocwind | StorageV2 | Standard_ZRS | rg-uks-prod-rovco-cwind | UK South |
| stukswatchguardfirebox | StorageV2 | Standard_GRS | rg-uks-watchguard-firebox-cloud-storage | UK South |
| ukscorediags | Storage | Standard_LRS | core-bootdiagstorageaccounts | UK South |
| uksservicesdiags | Storage | Standard_LRS | core-bootdiagstorageaccounts | UK South |
| ukwcorediags | Storage | Standard_LRS | core-bootdiagstorageaccounts | UK West |
| ukwservicesdiags | Storage | Standard_LRS | core-bootdiagstorageaccounts | UK West |
| vmbootup | Storage | Standard_LRS | rg-uks-wvd-it | UK South |
| vmrestorewvd | StorageV2 | Standard_RAGRS | rg-uks-prod-wvd-it | UK South |
| w06h5mgmsbackupasrcache | Storage | Standard_LRS | rg-sea-prod-singapore | UK South |


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| | | | | |
|---|---|---|---|---|
| watchgaadstorage | Storage | Standard_LRS | rg-uks-watchguard-firebox-cloud | UK South |
| watchguarddev | Storage | Standard_LRS | rg-uks-watchguard-dev | UK South |

### 6.2.2 Recommendations

| **Storage Account Recommendations** | |
|---|---|
| Recommendation | Rationale |
| Review and delete storage accounts no longer in use | Improves data security and reduces storage costs |
| Use General-purpose v2 storage accounts | Reduces cost in comparison to other Azure storage services |


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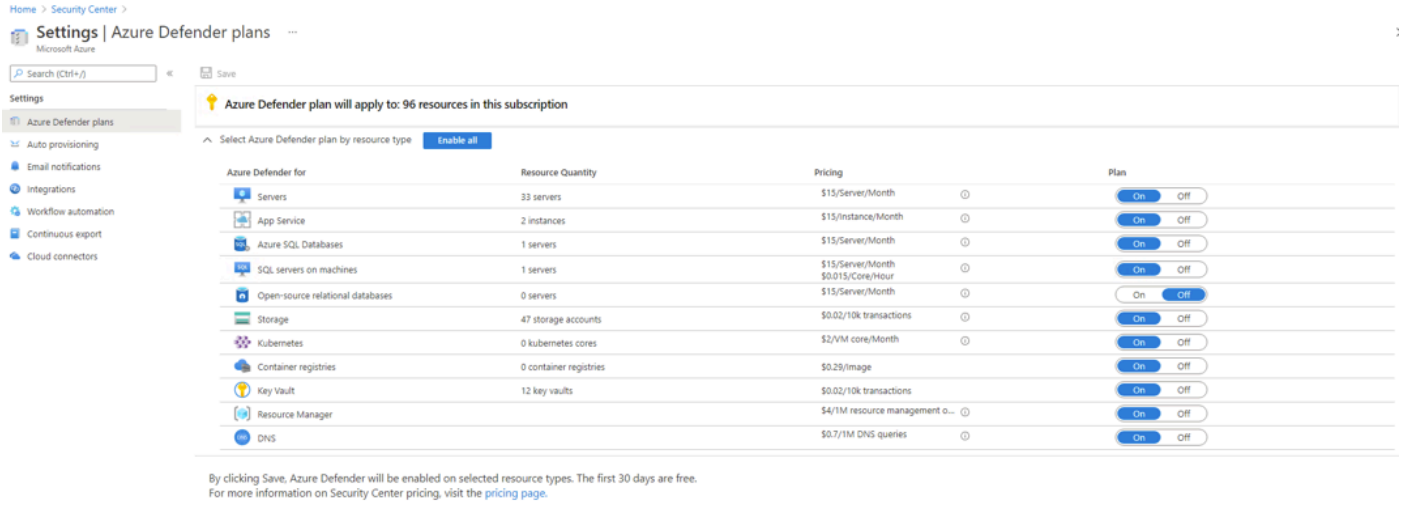
# 7 Security

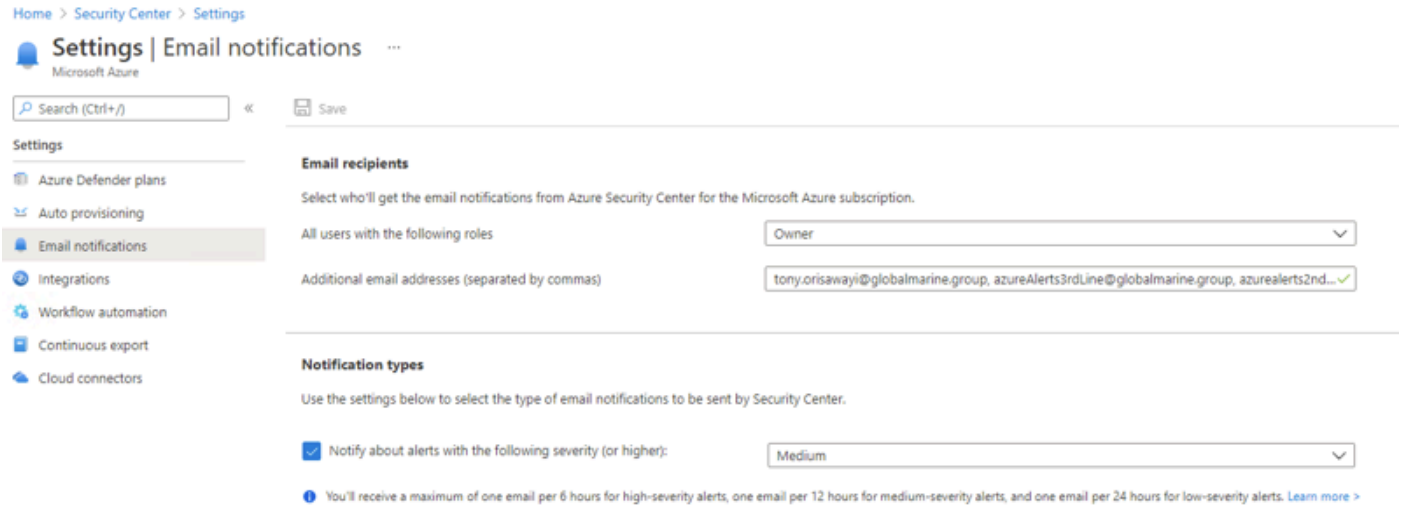
## 7.1 Azure Security Center

### 7.1.1 Current State

Azure Defender has been enabled for the following resources:




Security Center Email Notifications have been configured as follows:




The following Regulatory Compliance standards have been enabled: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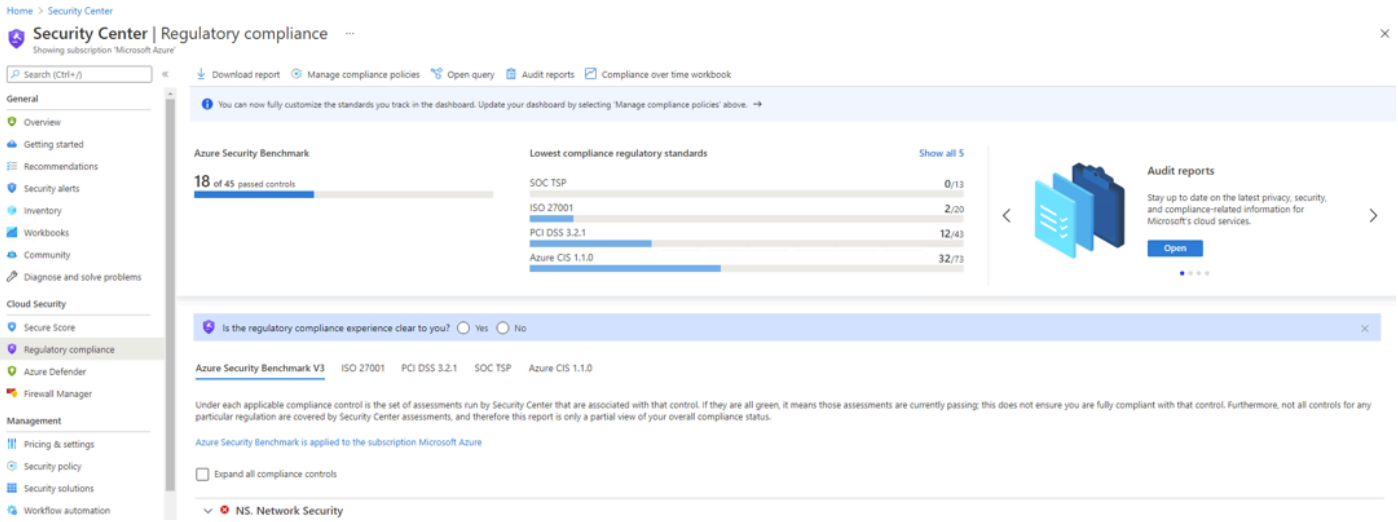


The following attachment contains a list of Azure Security Center alerts:

Azure Security Center Alerts.csv

The following attachment contains a list of Azure Security Center recommendations:

Azure Security Center Recommendations.csv

### 7.1.2 Recommendations

| Azure Security Center Recommendations |
|---|
| Recommendation |
| Enable Azure Defender for all resources |
| Enable automatic provisioning for Log Analytics agent for Azure VMs |
| Enable automatic provisioning for Vulnerability assessment for machines |
| Review and remediate Azure Security Center alerts prioritising high severity alerts |
| Review and remediate Azure Security Center recommendations prioritising high severity alerts |
| Review regulatory compliance requirements and disable any standards not required |

## 7.2 Azure Key Vault

### 7.2.1 Current State

| Key Vaults |
|---|


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| Name | Resource Group | Location |
|---|---|---|
| akv-2-7kgt26qkbaht4eb | rg-uks-wvd-oceaniq | UK South |
| AZ-SEA-MIGRATE1293kv | rg-sea-azure-migrate | Southeast Asia |
| kv-eus-storage | rg-eus-production | East US |
| kv-neu-storage | rg-uks-prod-akv | North Europe |
| KV-SEA-DISKS | rg-sea-prod-singapore | Southeast Asia |
| kv-sea-storage | rg-sea-akv | Southeast Asia |
| kv-uks-prod-akv01 | rg-uks-prod-akv | UK South |
| kv-uks-storage | rg-uks-prod-akv | UK South |
| kv-ukw-storage | rg-ukw-akv | UK West |
| kv-weu-storage | rg-uks-prod-akv | West Europe |
| UKS-DEV-O365-KV | rg-uks-dev-o365 | UK South |
| uks-vcentre1291kv | rg-sea-azure-migrate | UK West |

### 7.2.2 Recommendations

| **Key Vault Recommendations** | |
|---|---|
| Recommendation | Rationale |
| Create one Key Vault per application per environment | Separation of Key Vaults |
| Migrate vault access policies to RBAC | Azure RBAC key benefits over vault access policies:<br>- Provides a unified access control model for Azure resources by using the same API across Azure services<br>- Centralized access management for administrators - manage all Azure resources in one view<br>- Integrated with Privileged Identity Management for time-based access control<br>- Deny assignments - ability to exclude security principals at a particular scope. For information, see Understand Azure Deny Assignments |
| Restrict network access using Private Endpoints | Provides private and secure network connectivity by allocating a private IP address from a virtual network, effectively bringing Key Vault in to a virtual network |

## 7.3 Azure Update Management

### 7.3.1 Current State

| **Update Management** |
|---|
| Update Management has been enabled as a solution for one of the Log Analytics workspaces. However, VM's have not yet been onboarded to Update Management and deployment schedules have not been configured. |


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### 7.3.2 Recommendations

| Update Management |
| --- |
| Enable Update Management on each Log Analytics Workspace and Automation Account in each region. Configure deployment schedules according to uptime requirements and configure automatic onboarding by enabling automatic provisioning the Log Analytics agent to the same workspace in Security Center. |

## 7.4 Azure Remote Access

### 7.4.1 Current State

| Azure Remote Access |
| --- |
| The current method of remote administrative access to Azure VMs has not been formalised. |

### 7.4.2 Recommendations

| Azure Remote Access |
| --- |
| Implement Azure Bastion in the hub network to provide secure RDP/SSH access to Azure VMs using the Azure Portal. |


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# 8 Monitoring and Logging

Azure Monitor is a service in Azure that helps collect and analyse data that's generated by Azure resources. It gives real-time insights by using integrated search and custom dashboards by readily analysing records across all workloads. This helps in understanding how the applications are performing and proactively identifies issues affecting them and the resources they depend on.

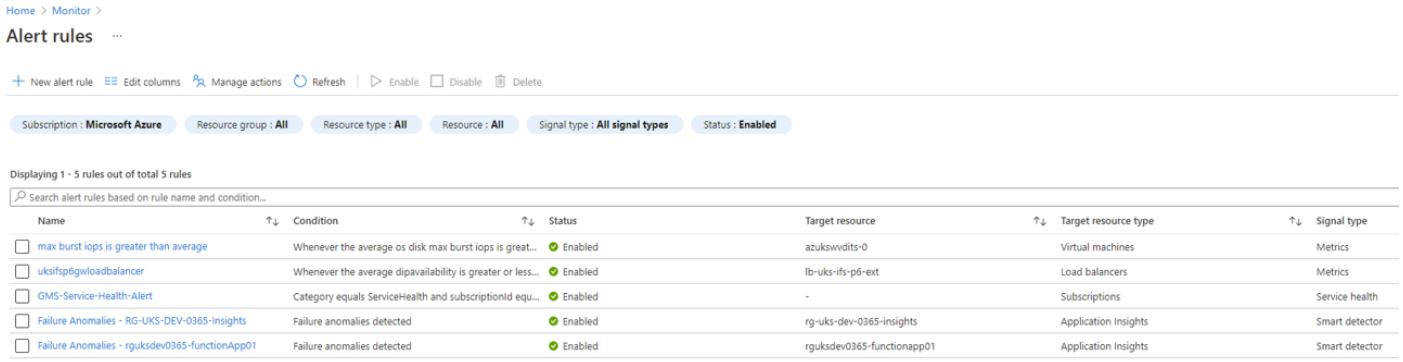
Azure Monitor collects data from multiple sources into a common data platform where it can be analysed for trends and anomalies. It differentiates between two data types:

- Metrics: Host level or guest level telemetry data gathered in near real time, e.g. CPU utilization.
- Logs: Log data produced by the resources, e.g. Windows event logs.

## 8.1 Azure Monitor

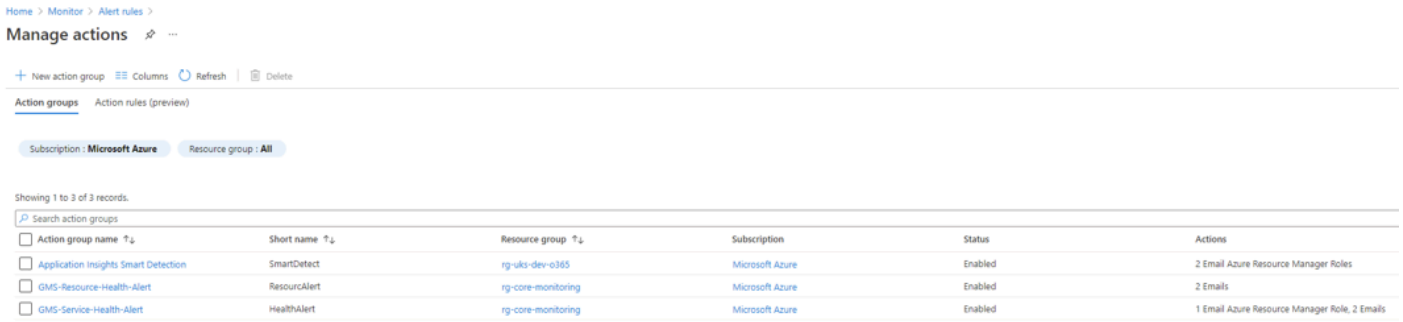
### 8.1.1 Current State

The following alert rules have been created in Azure Monitor:




| Name | Condition | Status | Target resource | Target resource type | Signal type |
|---|---|---|---|---|---|
| max burst iops is greater than average | Whenever the average os disk max burst iops is great... | Enabled | azukswvdits-0 | Virtual machines | Metrics |
| uksifsp6gwloadbalancer | Whenever the average dipavailability is greater or less... | Enabled | lb-uks-ifs-p6-ext | Load balancers | Metrics |
| GMS-Service-Health-Alert | Category equals ServiceHealth and subscriptionId equ... | Enabled | - | Subscriptions | Service health |
| Failure Anomalies - RG-UKS-DEV-0365-Insights | Failure anomalies detected | Enabled | rg-uks-dev-0365-insights | Application Insights | Smart detector |
| Failure Anomalies - rguksdev0365-functionApp01 | Failure anomalies detected | Enabled | rguksdev0365-functionapp01 | Application Insights | Smart detector |

The following actions groups have been created in Azure Monitor:




| Action group name | Short name | Resource group | Subscription | Status | Actions |
|---|---|---|---|---|---|
| Application Insights Smart Detection | SmartDetect | rg-uks-dev-o365 | Microsoft Azure | Enabled | 2 Email Azure Resource Manager Roles |
| GMS-Resource-Health-Alert | ResourcAlert | rg-core-monitoring | Microsoft Azure | Enabled | 2 Emails |
| GMS-Service-Health-Alert | HealthAlert | rg-core-monitoring | Microsoft Azure | Enabled | 1 Email Azure Resource Manager Role, 2 Emails |

### 8.1.2 Recommendations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| Azure Monitor Recommendations | |
|---|---|
| Recommendation | Rationale |
| Define monitoring requirements and create Alert Rules for each resource type. For example, the following Alert Rules can be configured for Virtual Machines:<br>- No Heartbeat alert<br>- Create or update virtual machine<br>- Percentage CPU (dynamic threshold)<br>- Disk Read Ops/Sec (dynamic threshold)<br>- Disk Write Ops/Sec (dynamic threshold)<br>- Memory | Alerts will provide notifications for Azure resources to allow issues to be identified and addressed proactively. |
| Create Azure dashboards to visualize monitoring data | Dashboards are useful to provide an organised view of resources and quickly launch tasks for day-to-day operations. For example, custom dashboards can be created based on projects, tasks, or user roles. |

## 8.2 Log Analytics Workspaces

### 8.2.1 Current State

| Log Analytics Workspaces | | | |
|---|---|---|---|
| Workspace Name | Resource Group | Location | Subscription |
| DefaultWorkspace-03e8c261-92f4-4a4c-a6ce-ad3ae17fd911-SEA | DefaultResourceGroup-SEA | Southeast Asia | Microsoft Azure |
| DefaultWorkspace-03e8c261-92f4-4a4c-a6ce-ad3ae17fd911-SUK | DefaultResourceGroup-SUK | UK South | Microsoft Azure |
| GlobalMarine | LogAnalyticsRG | UK South | Microsoft Azure |
| GMG-EUS-PROD-LogAnalytics | RG-EUS-PROD-LogAnalytics | East US | Microsoft Azure |
| gmg-sea-prod-loganalytics | rg-sea-prod-loganalytics | Southeast Asia | Microsoft Azure |
| GMG-UKS-PROD-LogAnalytics | RG-UKS-PROD-LogAnalytics | UK South | Microsoft Azure |
| RG-LA-OceanIQ-WVD-Monitor | RG-UKS-WVD-OceanIQ | UK South | Microsoft Azure |
| RG-UKS-DEV-0365-LogAnalytics | RG-UKS-DEV-O365 | UK South | Microsoft Azure |
| web-admin-portal-law-7kgt26qkbaht4 | RG-UKS-WVD-OceanIQ | UK South | Microsoft Azure |
| WVD-Monitor | RG-UKS-WVD-IT | UK South | Microsoft Azure |

### 8.2.2 Recommendations

| Log Analytics Workspace Recommendations |
|---|


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| Recommendation | Rationale |
|---|---|
| Consolidate multiple workspaces in each region in to a single workspace | It is recommended to create a single workspace per region due to the following benefits:<br>- All data resides in one place allowing for efficient correlation of data<br>- Simplified dashboard authoring, using Azure Workbooks, avoiding cross-workspace queries<br>- Having a workspace in the same region as resources will prevent additional ingress/egress bandwidth costs between Azure regions |
| Review and delete empty workspaces:<br>- GMG-EUS-PROD-LogAnalytics<br>- web-admin-portal-law<br>- WVD-Monitor | Prevents workspace sprawl and improves management of monitoring and logging capabilities |
| Review and configure log data retention period | Workspaces have 31 days of data retention by default at no cost. Variable data retention should be used if log data is needed for a longer time period, for example, 365 days of retention for Security and Audit logs. Retention can be configured at the workspace level for all logs, or at the table level for specific logs. |
| Review usage of daily volume caps | Daily caps can be configured to limit the daily ingestion for a workspace. However, these should be used with caution to prevent reaching the daily limit as this may result in loss of data for the remainder of the day and reduce the ability to observe and receive alerts. The daily cap is intended to be used as a way to manage an unexpected increase in data volume. |
| Review usage of Log Analytics Workspace Solutions:<br>- AzureAppGatewayAnalytics<br>- ChangeTracking<br>- NetworkMonitoring<br>- Security<br>- ServiceMap<br>- SQLAdvancedThreatProtection<br>- SQLVulnerabilityAssessment<br>- Updates<br>- VMInsights | Azure monitor offers a number of monitoring solutions which provide additional insight into the operation of a particular application or service. These solutions will store their data within a workspace and will affect ingestion costs. |

## 8.3 Platform Logs

### 8.3.1 Current State

| **Platform Logs** | |
|---|---|
| Setting | State |


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| | |
|---|---|
| Azure Active Directory Logs | Partially enabled – diagnostics settings enabled for Azure AD Tenant to send *AuditLogs*, *SignInLogs* and *ProvisioningLogs* to Log Analytics Workspace *GMG-UKS-Prod-LogAnalytics*. |
| Azure Resource Logs | Partially enabled – diagnostics settings enabled for some resources to send log data to Log Analytics workspace. |
| Azure Subscription Activity Log | Partially enabled – diagnostics settings enabled for Azure Subscription to send *Security* logs to Log Analytics Workspace *GMG-UKS-Prod-LogAnalytics*. |

### 8.3.2 Recommendations

| **Platform Logs Recommendations** | |
|---|---|
| Recommendation | Rationale |
| **Azure Active Directory Logs**: Update diagnostics setting to include all Azure AD logs. | Azure AD logs contains the history of sign-in activity and audit trail of changes made in the Azure Active Directory for a particular tenant. |
| **Azure Resource Logs**: Implement Azure Policies for each resource type which requires logging to be enabled. | Azure Resource Logs provide insight into operations that were performed within an Azure resource (the data plane), for example getting a secret from a Key Vault or making a request to a database. The content of resource logs varies by the Azure service and resource type. Deploying diagnostics settings using Azure Policy will ensure logging is enabled for existing and new resources. |
| **Azure Subscription Activity Logs:** Implement Azure Policy – *Configure all Azure Activity logs to stream to specified Log Analytics workspace*. | This policy will ensure all Azure Activity logs are streamed to the specified Log Analytics workspace. Azure Activity Logs provide insight into the operations on each Azure resource in the subscription from the outside (the management plane) in addition to updates on Service Health events. The Activity Log is used to determine the what, who, and when for any write operations (PUT, POST, DELETE) taken on the resources in your subscription. |


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# 9 Backup and Disaster Recovery

The Azure Backup service provides the ability to backup on-premises, Azure VMs, SQL servers/databases and other components to Microsoft Cloud.

Azure Site Recovery is a Disaster Recovery as a Service (DRaaS) offering that provides a simple tool to configure complex disaster recovery scenarios for workloads running on physical and virtual machines, either on-premises or on Azure, so that in the case of an outage these workloads are quickly failed over from a primary site to a secondary location.

## 9.1 Backup

### 9.1.1 Current State

| Recovery Services Vaults | | | | |
|---|---|---|---|---|
| Recovery Services Vault Name | Storage Redundancy | Resource Group | Location | Subscription |
| GMS-Backup-Vault2-UKS | GRS | core-networking | UK South | Microsoft Azure |
| GMS-Backup-Vault3-SEA | GRS | rg-sea-prod-singapore | Southeast Asia | Microsoft Azure |
| GMS0Backup-Vault3-SEA | GRS | core-networking | Southeast Asia | Microsoft Azure |
| Singapore-Servers-MigrateVault-khwcy8gs | GRS | rg-sea-azure-migrate | Southeast Asia | Microsoft Azure |
| uks-vcentre1291vault | GRS | rg-sea-azure-migrate | UK West | Microsoft Azure |

| Backup Policies | |
|---|---|
| GMS-Backup-Vault2-UKS Backup Policies | |
| Backup Policy Name | Backup Policy Type |
| DailyBackup | Azure File Share |
| DefaultPolicy | Azure Virtual Machine |
| HourlyLogBackup | SQL Server in Azure VM |
| GMS-D-8pm | Azure Virtual Machine |
| GMS-D-W-M-8pm | Azure Virtual Machine |
| GMS0Backup-Vault3-SEA Backup Policies | |
| Backup Policy Name | Backup Policy Type |
| DefaultPolicy | Azure Virtual Machine |
| HourlyLogBackup | SQL Server in Azure VM |
| GMS-D-8pm | Azure Virtual Machine |
| GMS-D-W-M-8pm | Azure Virtual Machine |

### 9.1.2 Recommendations

| Backup Recommendations |
|---|


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| Recommendation | Rationale |
| --- | --- |
| Configure diagnostics settings on each Recovery Services Vault via Azure Policy to send data to Log Analytics workspace https://docs.microsoft.com/en-us/azure/backup/configure-reports | Diagnostics settings are required in order to view Azure Backup Reports.<br><br>Azure Backup Reports provide insights on backups based on data that spans a long period of time. Use cases for such a solution include:<br>- Allocating and forecasting of cloud storage consumed.<br>- Auditing of backups and restores.<br>- Identifying key trends at different levels of granularity. |
| Use built-in RBAC roles to control backup management operations:<br>- Backup Contributor<br>- Backup Operator<br>- Backup Reader | Using the built-in RBAC roles ensures adherence to the principle of least privileges for just enough administration. |
| Consider implementing Cross Region Restore to allow Azure VMs to be restored to a secondary, paired region. | This option allows an organisation to meet audit or compliance requirements, and to restore the VM or its disk if there's a disaster in the primary region. |
| Consider implementing Auto-Enable Backup on VM Creation using Azure Policy. https://docs.microsoft.com/en-us/azure/backup/backup-azure-auto-enable-backup | Automatic enablement of VM backups in an organization will ensure that all business-critical machines are backed up with the appropriate retention. |

## 9.2 Disaster Recovery

### 9.2.1 Current State

| **Disaster Recovery** |
| --- |
| Azure Site Recovery has not been implemented for the Azure Subscription. |

### 9.2.2 Recommendations

| **Disaster Recovery** |
| --- |
| Azure Site Recovery should be considered as part of the overall BCDR strategy to provide resilience against regional Azure issues. |


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# 11 Appendix

## 11.1 Azure Resources List

The following attachment contains a list of all resources exported from the Azure Portal:

Azure Resources List.csv

## 11.2 Summary of Recommendations

The following attachment contains a summarised list of all recommendations:

Summary%20of%20Azure%20Health%20Ch

## 11.3 Service Improvement Plan

The following attachment contains a prioritised list of service improvement plans:

Azure%20Service%20Improvement%20Plan


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# 12 Glossary of Terms

| Abbreviation | Description |
|---|---|
| API | Application Programming Interface |
| ARM | Azure Resource Manager |
| Azure AD | Azure Active Directory |
| BaaS | Backup as a Service |
| DNS | Domain Name Services |
| DRaaS | Disaster Recovery as a Service |
| GRS | Geo-redundant Storage |
| IaC | Infrastructure as Code |
| IDaaS | Identity as a Service |
| JIT | Just in Time access |
| LRS | Locally-redundant Storage |
| MFA | Multi Factor Authentication |
| MG | Management Group |
| MMA | Microsoft Monitoring Agent |
| NSG | Network Security Groups |
| RBAC | Role Based Access Control |
| RG | Resource Group(s) |
| SAML | Security Assertion Mark-up Language |
| SSL | Secure Socket Layer |
| SSD<br>HDD | Solid-State Drive<br>Hard Disk Drive |
| TLS | Transport Layer Security |
| UDR | User Defined Routes |
| VM | Virtual Machine |
| VNet | Virtual Network |
| vWAN | Virtual Wide Area Network |
| VPN | Virtual Private Network |


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